Version of Record: https://www.sciencedirect.com/science/article/pii/S037842662100131X
Manuscript_800497eebf5b9291c2d6ede410112633

Risk-adjusted Return Managed Carry Trade

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We propose a simple dynamic currency carry trade strategy. Implementation of this strategy requires only two indicators: the aggregate forward discount and historic, global foreign exchange volatility. We find that our dynamic strategy delivers a significantly higher Sharpe ratio and greater skewness than a benchmark carry trade strategy. Our results are robust to the time period and currency choices. Both indicators contribute to predictability, and the improvement in performance is visible in both good and bad times. Standard risk factor models are unable to fully explain the returns of this augmented carry trade.

JEL Classification numbers: F31, G12

Keywords: Exchange Rate, Carry Trade, Portfolio, Risk premium

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1 Introduction

The currency carry trade is a popular investment strategy in which investors buy high-yielding currencies and sell low-yielding ones. On average, high-yielding currencies tend to exhibit a positive return, while low-yielding currencies tend to exhibit a negative return in contrast to the prediction of the Uncovered Interest rate Parity (UIP). However, like many other strategies, the carry trade exhibits substantial time variation in both risk and return (Figure 1), suggesting that profitability can potentially be enhanced by timing the exposure to the strategy. Investors in particular might wish to increase their exposure (i.e., to take more risk) when the strategy's expected risk-return trade-off is favorable.

In this paper, we show that the conditional risk-return trade-off can be approximated by the time-varying average forward discount (F/S) of the currencies, especially those that are attractive to investors because their costs of trading do not outweigh the discount (e.g., Burnside et al., 2007). Our representative investor takes greater risk when the average forward discount of the attractive currencies is greater and takes less risk when it is smaller.

In Figure (2), we provide the intuition for our proposition by grouping monthly carry trade returns by the previous month's average forward discount of the attractive currencies (Figure 2.a). We see that the profitability of the carry trade is higher when the average forward discount is greater during the preceding period, which suggests that investors can benefit from forward discount timing.
Following Moreira and Muir (2017), we also group monthly carry trade returns using the previous month's realized volatility and plot average returns and volatility over the subsequent month (Figure 2.b). We confirm their observations from the equity market; the relationship between lagged volatility and average returns is weak, but there is a strong relationship between lagged volatility and current volatility (2.c), thereby suggesting that investors can also benefit from volatility timing (Figure 2.d).

Based on these insights, we construct augmented strategies that rely on the forward discount timing and volatility timing and find that both signals improve the Sharpe ratio of the carry trade to a comparable extent. While our benchmark equally weighted carry trade strategy (i.e. HML-FX of Lustig, Roussanov and Verdelhan, 2011) has a Sharpe ratio of 0.76 in our sample, the forward discount adjusted version of the strategy exhibits a Sharpe ratio of 0.99, and the volatility-adjusted version exhibits a Sharpe ratio of 0.84. Interestingly, we find that the combination of these signals

in a single indicator further improves the return to the carry trade up to 1.07. A rolling Sharpe ratio shows that the augmented carry trade can benefit from timing exposure in both good and bad times on average, turning the skewness of the distribution of returns from negative (-0.76) to positive (0.97). Also, timing exposure tends to lower the impact of the drawdowns observed during bad times.

We document that the abnormal returns of our augmented carry trade tend to persist even when we control for popular risk factors extracted from the equity, bond, or FX markets. Abnormal returns remain especially significant and economically meaningful when we consider Lustig et al. (2011) HML-FX carry as a factor. This finding suggests that the return to our strategy does not take profit from exposure to traditional risk factors and that the carry trade remains, at least partially, a puzzle[1] (e.g. Jorda and Taylor, 2012, Barroso and Santa-Clara, 2015b, Daniel et al., 2017, Sokolovski, 2019, Bekaert and Panayotov, 2020, among others).

According to our results, the augmented strategy actually exploits the instability of UIP, which tends to hold during some periods but not in others (see Bekaert and Hodrick, 2018). In particular, large values of our indicator are associated with negative values of the Fama regression coefficient (consistent with positive carry trade returns). This observation echoes the conclusions of Clarida, Davis, and Pedersen (2009), who establish a link between the up and down regimes of the coefficient of the Fama regression and volatility. We extend this conclusion to the value of the average forward discount attractive to investors. Our observations suggest that the lower betas of the Fama regression are associated with the higher values of the forward discount and vice versa.

We obtain our main results in a large cross-section of 34 currencies covering the period from 1999 to 2018. We also provide results for a sample of 39 currencies covering the period from 1983 to 2018 and a sample of 40 currencies covering the period from 1976 to 2018. For G10 currencies, the augmented strategy only exhibits a small, insignificant increase in the Sharpe ratio from 0.45 to 0.53 when the signal is strictly extracted from this sample. However, when we include information from EM currencies in the signal, the Sharpe ratio increases to 0.64, and the skewness goes again from negative (-0.50) to positive (0.17).

We study various carry trade strategies, notably the well-known equally weighted version similar to the Lustig et al. (2011) HML-FX factor and the discount forward-weighted version of Daniel, Hodrick, and Lu (2017). The results are almost independent of the way we build the strategy and are confirmed by several robustness tests. In particular, we control for various specifications of the signal, for outliers in the series, for idiosyncratic risk, and for a change in the numeraire.

Our conclusion remains unchanged regarding whether we consider full transaction costs available in our data, roll-over transaction costs or zero-transaction costs. Also, we show that the typical investor can benefit from carry trade timing, even if one is subject to hard leverage constraints. Investors preferring positive skewness would favor the managed version of the strategy, even when the benchmark strategy is optimized in a standard fashion.

Several papers have already related successfully observed forward discounts or volatility and future returns of the carry trade. Asness et al. (2013), Jurek (2014) and Daniel, Hodrick, and Lu (2017) relate the observed forward discount and future returns of the carry trade in the cross section. They conclude that the spread weighted portfolios with a constant amount invested over time tend to outperform the benchmark equally weighted portfolios.
Another strand of the literature considers carry trade investors' leverage and volatility. For instance, Bakshi and Panayotov (2013) control the amount invested in the carry trade according to the prediction of their model, which is based on realized volatility, changes in the price of a commodity index, and changes in the measure of market liquidity. They consider binary positions; they invest the same amount in the strategy when their model indicates future positive returns and reduce it to zero for other occasions. Cenedese, Sarno, and Tsiakas (2014) study the same type of binary investment. They invest in the carry trade except during the months that follow periods of high realized volatility and low realized returns, defined by certain thresholds. Bhansali (2007) and Melvin, Prins, and Shand (2013) also use volatility to time investment in the carry trade, while Jorda and Taylor (2012) control their investment using deviations from the Purchasing Power Parity, among other indicators.

In a similar work, Mulder and Tims (2018) adjust the set of investable currencies when volatility is high. Bekaert and Panayotov (2020) also propose a strategy in which they routinely exclude, from the set of investable currencies, the currencies that impair the ex-post Sharpe ratio of the strategy in the preceding period. Finally, Della Corte, Ramadorai, and Sarno (2016) propose a strategy based on idiosyncratic implied volatility weightings that produces larger Sharpe Ratios than the carry trade itself. Egbers and Swinkels (2015), by contrast, conclude that volatility does not act as a significant predictor of the future returns of the carry trade. However, their work differs from the other aforementioned articles, as they use implied volatility or a change in implied volatility extracted from both the equity and the foreign exchange markets to enter and exit the currency carry trade (and not realized volatility).

In this paper, we scale investments continuously (not in a binary fashion), based on a risk-

adjusted return signal that combines both forward-discount timing and volatility timing. Our inspiration comes notably from Barroso and Santa Clara (2015a) and Moreira and Muir (2017), who show the importance of volatility timing. We add to this literature by showing that forward discount timing (E) produces very similar results to volatility timing (1/V) for the carry trade. More importantly, we show that the combination of both partial signals in a unique indicator (E/V) further improves the return of the strategy. As a result, investors might favor this (E/V) indicator over simple (E) or (1/V) signals. We also take inspiration from Daniel and Moskowitz (2016), who dynamically adjust the leverage of the momentum strategy in the equity market using forecasts of returns and variances. However, unlike their approach, we only use past observations (no estimation − no forecast).

Recently, Maurer, Pezzo, and Taylor (2019) and Maurer, To, and Tran (2020) show that controlling for time-varying exposure (notional value) by means of forward discounts and volatility enhances the profitability of optimized portfolios of carry trade. Ackermann, Pohl, and Schmedders (2016) and Daniel, Hodrick, and Lu (2017) also worked on optimized portfolios. In this article, we show that comparable results can be obtained for the well-known non-optimized, equally weighted portfolio (HML-FX type), especially when taking transaction costs in consideration. Our results suggest that market timing rather than portfolio construction is the primary source of profitability enhancement. This is important because the HML-FX type carry trade can be easily reproduced by investors. In particular, investors do not have to worry about possible errors in the covariance matrix estimation, which is a well-known limit to portfolio optimization (e.g., Baz et al., 2001; Brandt, 2005; Della Corte et al., 2009; Laborda et al., 2014; Barroso and Santa-Clara, 2015b).

Our strategy works particularly well in the post-euro era and when we include emerging and frontier markets. It complements the findings of Maurer, Pezzo, and Taylor (2019) and Maurer, To, and Tran (2020) who mention that they tend to obtain weaker results, respectively, when they consider emerging markets' currencies and post-1999 data. For many years, the literature on portfolio construction compared naive strategies like ours to optimized ones. For instance, De Migguel et al. (2009) claim that naive equally weighted strategies tend to do better than optimized strategies. Our objective in this paper is not to fuel this debate. Yet, we point out that leverage investors can more easily implement naive portfolios than optimized ones.

The rest of the paper is organized as follows: in Section two, we detail the construction of the risk-adjusted return signal. In Section three, we present the data and report the empirical findings. We provide robustness tests and additional results in the fourth section. In Section five,

we provide complementary tests that, notably, take into account investors' preference for skewness. In Section six, we look at the relationship between our indicator and the time-varying beta in the Fama regression. We also relate the augmented strategy to factor models. Finally, we conclude in the seventh section.

2 The Signal

To build our signal, we rely on what Burnside et al. (2007) label the rationale for currency speculation. We use f_t^b and f_t^a as well as s_t^b and s_t^a to denote the bid price and ask price of the forward and spot exchange rate of the Foreign Currencies (USD/FCU meaning 1 USD = x FCU) at time t. If agents think $1/s_{t+1}^b$ and $1/s_{t+1}^a$ are random walks[2], then their decision rule is as follows:

$$\begin{cases} \text{Buy the FCU if} & E(f_t^b/s_{t+1}^a) = f_t^b/s_t^a > 1 \\ \text{Sell the FCU if} & E(f_t^a/s_{t+1}^b) = f_t^a/s_t^b < 1 \\ \text{No Transaction if} & f_t^a/s_t^b > 1 \text{ and } f_t^b/s_t^a < 1 \end{cases} \tag{1}$$

This decision rule defines a transaction region in which investors buy and sell currencies and a no-transaction region in which they do not invest (e.g., Liu, 2004). Burnside et al. (2007) and Barroso and Santa Clara (2015) rely on this definition to define precisely the currencies that are attractive to the investors (transaction region) and the currencies that are not attractive to the investors (no-transaction region).

Then, in t, the total expected profit π_{t+1} of investing in the carry trade for a position of one USD in the set of attractive investment currencies and one dollar in the set of attractive financing currencies is:

$$E_t(\pi_{t+1}) = \left[\sum_i (f_t^b/s_t^a)_i - 1 \mid (f_t^b/s_t^a)_i > 1\right] / n_t^{buy} - \left[\sum_j (f_t^a/s_t^b)_j - 1 \mid (f_t^a/s_t^b)_j < 1\right] / n_t^{sell} \tag{2}$$

where n_t^{buy} and n_t^{sell} are respectively the number of investment and financing currencies attractive at time t[3]. This measure of the expected profit can be understood as the difference between

the average mass of forward discount (premium) per attractive currency. Our main contribution to the literature is to use it as a scaling signal for investing in the carry trade strategy. Large values of the expected profit are especially associated with large values for the leverage of the strategy and vice versa.

Recently, Della Corte, Ramadorai, and Sarno (2013); Cenedese et al. (2014); Ackermann, Pohl, and Schmedders (2016); Daniel, Hodrick, and Lu (2017); Moreira and Muir (2017); Maurer, Pezzo, and Taylor (2019); and Maurer, To, and Tran (2020) show that controlling for volatility is key to time market returns in the foreign exchange market. Therefore, we complement our indicator by also including the information content of the contemporaneous level of FX market variance (MV). This shapes a mean variance indicator, which differs from those based either on constant expected returns (e.g., Moreira and Muir 2017), binary rules (e.g., Cenedese et al., 2014), forecasted returns (e.g., Daniel and Moskowitz, 2016), or optimized portfolios (e.g., Ackermann, Pohl, and Schmedders, 2016; Maurer, Pezzo, and Taylor, 2019 and Maurer, To, and Tran, 2020). In this paper, we show that the partial signal based on forward discounts conveys as much information as the partial signal based on volatility. Beyond this, the combination of both signals in a unique indicator further improves the statistics of the carry trade.

We capture the aggregate variance in FX by estimating the volatility of the excess return to the FX market portfolio, defined as the equally weighted portfolio of all currencies in the universe of investment. This can be thought of as the excess return to a naive $1/N$ currency trading strategy, in which the return would be $r_{Mt} = \frac{1}{N}\sum_i f_{i,t}/s_{i,t+1} - 1$. Comparable conclusions can be reached using the volatility of the carry trade portfolio itself.

A simple and popular way of estimating a realized measure of monthly market variance (MV) is to use the daily excess returns as follows:

$$MV_t = \sum_{d=1}^{D_t} r_{Mt,d}^2/D_t + 2\sum_{d=2}^{D_t} r_{Mt,d}/D_t * r_{Mt,d-1}/D_t \tag{3}$$

where D_t is the number of trading days in month t (typically $D_t = 21$). This measure of market variance accounts for the autocorrelation in daily returns (e.g., French, Schwert, and Stambaugh, 1987; Goyal and Santa-Clara, 2003; and Bali, Cakici, Yan, and Zhang, 2005, among others).

Finally, we define our indicator as:

$$Signal_t = \frac{E_t(\pi_{t+1})}{(MV_t)} \tag{4}$$

This indicator (Figure 3) can be understood as a forward-looking signal (expected returns) to noise (volatility) ratio of the carry trade. In the currency market, part of the risk premium is known in advance (i.e. the forward discount). Therefore, it seems natural for investors to scale their leverage on the back of the information they hold about both the risk and the premium attached to the risk. An investor who maximizes the Sharpe ratio of the benchmark carry trade by dynamically scaling investments would consider this signal as the optimal leverage at time t (e.g., Daniel and Moskowitz, 2016).

For a given month t, the signal is typically the ratio of the expected profit (proxied by the average forward discount of the attractive currencies) observed at the end of month t to hold a position from the end of month t to the end of month $t+1$ to the observed volatility during month t. Note, however, that the volatility indicator captures market volatility, not just the volatility of the currencies attractive to the investor. To ensure no overlap between the signal and the future return, we further lag the signal by one month. Therefore, the signal obtained at the end of month t is used for the positions held from the end of month $t+1$ to the end of month $t+2$.

In this paper, we study the predictive power of this indicator for investment in the carry trade during the next period. We especially expect to observe a positive correlation between the average discount adjusted by the volatility in one period and the return of the carry trade in the future.

3 Data Analysis

We use data collected by Barclays and Reuters that is available on Datastream for 39 different countries (i.e. currencies): Australia, Brazil, Belgium, Bulgaria, Canada, Chile, Colombia, Croatia, Czech Republic, Denmark, Euro Area, France, Germany, Hong Kong, Hungary, India, Israel, Italy, Japan, Kuwait, Mexico, the Netherlands, New Zealand, Norway, the Philippines, Poland, Russia, Saudi Arabia, Singapore, Slovakia, South Africa, South Korea, Sweden, Switzerland, Taiwan, Thailand, Turkey[4], United Arab Emirates, and the UK. The euro series starts in January 1999, replacing the German DEM. The maximum number of currencies in the largest dataset is 34. While forward rates might be available for a larger collection of currencies, many would have had virtually no liquidity (2019 *BIS Triennial Survey*). The data cover the period from November 1983 to July 2018. To take into account any possible measurement errors, we carefully clean[5] the data following Darvas (2009).

For our main test, we focus on post-1999 data. These data account for more than 75% of the full sample. We include pre-1999 data later for robustness tests. Indeed, pre-1999 data include

a period of unstable country risk (Mexico, Asian countries), characterised by CIP violations for certain currencies, as mentioned in Lustig et al. (2011). Bekaert and Hodrick (2018) refer to this period as the rocky 1990s. Also, before 1997, observations for emerging markets are scarce (we have data only for Hong-Kong, Singapore, and South Africa), which limits the possibility of diversification and could increase estimation errors. Finally, the dataset includes several pegged currencies (Hong-Kong and Singapore) and quasi-pegged converging currencies (Belgium, France, Italy, and the Netherlands) with possible peso problems. These pegged currencies account for up to 35% of the sample before 1997. Therefore, we present our main results in the sample covering the period from 1999 to 2018 and robustness tests in larger samples and in a sub-sample of G10 countries: Australia, Belgium, Canada, Denmark, Euro Area (Germany before 1999), France, Italy, Japan, Netherland, New Zealand, Norway, Sweden, Switzerland, and the UK over the period from November 1983 to July 2018.

3.1 Building currency portfolios

We compute the monthly excess return r^i_{t+1} observed in $t+1$ for holding any new position in a foreign currency i using log prices[6] as $r^i_{t+1} = f^i_t - s^i_{t+1}$ with f^i_t as the log forward rate of currency i observed at time t and s^i_{t+1} the log spot exchange rate observed at time t+1.

At the end of each month, we rank the currencies from low to high interest rate currencies on the basis of their forward discount, observed at that time[7]. The currency with the lowest forward discount receives rank 1, while the currency with higher forward discount receives rank N. The carry trade consists of buying (selling) forward high (low) yielding foreign currencies at time t and selling (buying) them back at the prevailing spot exchange rate at $t+1$. In practice, positions are not opened and closed every month. At a lower cost, investors can roll over the existing positions they wish to keep. We adjust the cost of trading the existing positions as proposed by Darvas (2009). In this case, accounting for bid-ask spread transaction costs (with prices denoted b or a as appropriate), the payoff of the carry trade is:[8]

$$\begin{cases} r^{long}_{t+1} = \frac{A^{new}_t * (f^b_t - s^a_{t+1}) + A^{rol}_t * (f^b_t - s^a_{t+1} + (s^a_t - s^b_t))}{A^{new}_t + A^{rol}_t} & \text{if the investor buys the FCU} \\ r^{short}_{t+1} = \frac{A^{new}_t * (f^a_t - s^b_{t+1}) + A^{rol}_t * (f^a_t - s^b_{t+1} - (s^a_t - s^b_t))}{A^{new}_t + A^{rol}_t} & \text{if the investor sells the FCU} \end{cases} \tag{5}$$

with A^{new}_t and A^{rol}_t the notional amount, in USD, of new and rolled-over positions.

Currency portfolios. In this paper, we study the most common way to invest in the carry trade: we form equally weighted portfolios of currencies. This strategy is largely comparable to HML-FX, as developed by Lustig et al. (2011). We control the leverage by the signal defined in equation (4). Portfolios are rebalanced at the end of every period. When the currencies are ranked by their forward discount from 1 (lowest) to N (highest), the equally weighted portfolios of n carry-trade positions can be defined as:

$$\begin{cases} Buy \quad \mathrm{w}_i = 1 * signal \quad \text{if} \quad N - n + 1 < Rank_{f_t/s_t} < N, \\ \\ Sell \quad \mathrm{w}_j = -1 * signal \quad \text{if} \quad 1 < Rank_{f_t/s_t} < n, \end{cases} \tag{6}$$

Typically, when n=4, the investor holds an HML-FX type portfolio of four long positions in high-yielding currencies and four short positions in low-yielding ones. This strategy is symmetric both in number of currencies and value, as we invest $1 * signal$ in each four higher yielding currencies and $-1 * signal$ in each of the four lower yielding currencies. This strategy is dollar neutral, as the position in USD is implicitly set at zero but not numeraire independent, because if we define it relative to another benchmark currency, then the position in dollars would receive non-zero weights (see Bekaert and Panayotov, 2020 for a discussion). As the carry trade we study is a zero-investment and self-financing strategy, in theory, we can scale it with our signal without constraint by multiplying the weights by the signal. However, for real investors, risk control and margin requirements limit this possibility. We will demonstrate later that these limits are not binding for our conclusions. Also, note that the benchmark unleveraged carry trade is obtained when the signal is fixed at 1.

Finally, the notional amounts in USD A_t^{new} and A_t^{rol} can be defined as:

$$\begin{cases} Buy \quad A_t^{new} = max(0, w_{it} - w_{it-1}) \quad A_t^{rol} = min(w_{it}, w_{it-1}) \\ \\ Sell \quad A_t^{new} = min(0, w_{jt} - w_{jt-1}) \quad A_t^{rol} = max(w_{it}, w_{it-1}) \end{cases} \tag{7}$$

For the robustness tests of Section 4, we also build portfolios in which the n currencies are

weighted according to their forward discount, as in Asness et al. (2013) and Daniel, Hodrick, and Lu (2017). We call this strategy $Carry_{size}$. Finally, we also consider portfolios in which both the numbers of investment and financing currencies are time-varying according to the number of currencies which are attractive. We call this strategy $Carry_{ba}$ (see Equation (1) and Appendix A for a precise definition). However, for the rest of this paper, the benchmark carry trade is the equally weighted one defined by Equations (6) and (7).

The return to currency speculation. The returns to the portfolio of long positions and the portfolio of short positions are:

$$\begin{cases} R_{t+1}^{long} = \frac{1}{(nl)} \sum_{i=1}^{nl} w_i * r_{t+1}^i \\ \\ R_{t+1}^{short} = \frac{1}{(ns)} \sum_{j=1}^{ns} w_j * r_{t+1}^j \end{cases} \tag{8}$$

with nl as the number of currencies in which the investor has a long position and ns as the number of currencies in which it has a short position. Note, again, that our benchmark strategy is equally weighted $nl = ns$ (see Appendix A for the calculation of the returns of other types of carry trade). Finally, with $w_i > 0$ and $w_j < 0$, the return of the carry trade strategy is the sum of R_t^{long} *and* R_t^{short}.

3.2 Empirical findings

In Table I, we report the summary statistics for $Carry_n$ when $nl = ns = 4$ for both the unleveraged carry trade (constant signal set at 1) and the augmented version of the strategy when the signal is time varying[9]. We report the results in the main sample of 34 currencies for the period from January 1999 to July 2018. The results in the full sample including all data from 1983 to 2018 are largely similar and do not change the main contribution of this paper. To avoid the possible impact of small, illiquid currencies, we also report the results for G10 currencies. These results are reported in Section 4.4 and in the online appendix.

For every version of the strategy, we report the annualized mean, standard deviation, skewness, and kurtosis of the distribution of returns, the annualized Sharpe Ratio (SR), the annualized Information Ratio (IR) against the benchmark strategy, the annualized Sortino Ratio (SoR), and finally the annualized ratios of the mean return to the mean drawdown (M_{DD}) and maximum drawdown (Max_{DD}). The definition of these ratios can be found in Appendix B. We also report

the non-annualized version of these ratios and their standard errors obtained by bootstrapping the statistics[10]. Column p-v reports the p-value of the test of equality of the Sharpe ratios against the benchmark strategy of Ledoit and Wolf (2008). Each month, our managed strategy increases or decreases leverage according to the variation in our signal. For ease of interpretation, we set the average exposure of the managed strategy at one, in line with the unconditional exposure of the benchmark strategy. In this case, the exposure of the managed strategy varies from 0.25 to 2.7. As the benchmark strategy implies investing two dollars, the managed strategy invests an amount varying from 0.5 dollar to 5.4 dollars. Finally, we follow Moreira and Muir (2017), who present the results for portfolios with the same unconditional standard deviation as the benchmark portfolio. In our case, these standard deviations are very close in value, and they change the amounts invested very marginally. However, it is important to note that one needs sample information for such a calibration (e.g. Liu, Tang and Zhou, 2019 and Cederburg, O'Doherty, Wang and Yan, 2020). Therefore, in this form, the strategy is not directly implementable by real-time investors who would rely on backtesting to calibrate the optimal leverage at inception.

3.2.1 Main results

First, we confirm that our indicator can predict future currency returns. Indeed, when we regress the return to the benchmark carry trade on the signal, we find a positive beta of 0.018, which is significant at the 5% level. Higher values of the signal tend to indicate future positive returns, and vice versa. This makes sense because we know that the return to the currency carry trade comes from the interest rate component of the bet (e.g. Hassan and Mano, 2019 and Colacito, Riddiough, and Sarno, 2019). For instance, in our sample, the Sharpe ratio of the carry trade is negative if we ignore the interest rate component of the bet against positive when we include it. Logically, higher average forward discounts in a low volatility environment call for larger future returns. The R^2 of the regression only reaches 5%, but we suspect that this small predictive power is enough to control the leverage of the strategy. In our samples, the predictive power comes both from the average discount forward and the market volatility, especially for well-known currency carry trade like CHF, JPY, and ZAR (see the online Appendix A1a for more results).

When we look at the performances of the strategy in Table I, we see that, in our sample, the benchmark carry trade exhibits a Sharpe ratio of 0.76. This is equivalent to a non-annualized ratio of 0.22, which is significantly different from zero when we consider the value of the standard errors (0.07). However, Figure 1 shows that, after several years of high returns in the mid-2000s, the return to the carry trade strategy has been poor since 2008 and the decrease of interest rates in

the world.

Using our indicator to time the investment in the carry trade (see Figure 3a), we generate a higher risk-adjusted return: the signal pushes the Sharpe Ratio of the augmented strategy to almost 1.08. This represents a significant improvement (see Figure 3.b). Indeed, the test on the equality of the Sharpe Ratios as proposed by Ledoit and Wolf (2008) rejects equality with a p-value of 0.01. Also, the augmented strategy exhibits a positive skewness, which might be appealing for investors as it indicates a reduction of downside risk. This observation echoes the results of Jorda and Taylor (2012), Maurer, Pezzo, and Taylor (2019) and Maurer, To, and Tran (2020), who also obtain positive skewness for their augmented carry trade.

When we break down our comprehensive indicator into a partial signal based on the expected returns ($E(\pi_{t+1})$ of Equation 4) and another one based on market variance ($\frac{1}{(MV_t)}$ of Equation 4), we see that the contributions of these partial signals are similar. Indeed, when we use the average discount as the unique signal for investment (see Figure 3.c), the Sharpe Ratio reaches 0.99 against 0.76 for the benchmark carry trade. The improvement is large but not enough to be significant (p-v 0.06). When we use the market variance as a signal, the Sharpe ratio increases to 0.84 (see Figure 3.d) with a p-v of 0.31. This is an important result: we could obtain comparable Sharpe ratios by timing volatility or expected returns. This appends the conclusions of Moreira and Muir (2017), who only consider volatility timing but not expected returns timing for the carry trade. In addition, we show that combining both partial signals in the comprehensive indicator ($Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$) pushes the Sharpe ratio of the strategy higher (1.08). The other risk-adjusted returns convey the same information. As a result, it seems that investors should favor this (E/V) indicator over simple (E) or (1/V) signals.

Table I about here

The data available on Datastream are only indicative quotes for which the bid ask spreads might be somewhat inflated compared to real trades. Hence, one might be concerned that these bigger spreads distort the set of attractive currencies by excluding trades that might be marginally profitable in real data. However, traders facing volatile signals (Equation 1) might not engage in such trades automatically. This is in line with the assumption of Bacchetta and Van Wincoop (2010) that investors' portfolios of foreign exchange positions are not rebalanced continuously. Also, Karnaukh, Ranaldo, and Soderlind (2015) report that, among the larger bid ask spread

changes observed on Datastream, the ones that trigger the main revisions of the portfolios are consistent with real trades, as visible on EBS. Finally, in the online appendix to this paper, we report the return to the augmented strategy using full round-trip costs and mid-market quotes (no cost). The Sharpe ratio of the augmented strategy decreases from 1.21 (no cost) to 0.82 when we include full round-trip costs (1.08 with roll-over fees), but the observed improvement from the benchmark version remains significant in all cases (see Table V and VI in the online Appendix A2).

3.2.2 Performance attribution

To highlight the source of improvement, in Table II we report the summary statistics for both the investment leg and the financing leg of every version of the strategy. We see that the investment leg of the benchmark strategy has a Sharpe ratio of 0.62, while the financing leg exhibits a Sharpe ratio of 0.09. Interestingly, we see that the improvement observed for the augmented version comes from the change of the Sharpe ratio of the long leg (p-v 0.03), rather than from a change for the short leg (p-v 0.30). Notably, this improvement seems to come from the reduction of the tail risk of this long leg during bad times, as measured by the return to average and max drawdown ratios. We will see later that, in the sample covering the period 1983 to 2018, the improvement tends to come from the short leg of the strategy but still exhibits a reduction of the tail risk. This would tend to indicate that our signal can help to forecast large negative returns of the investment currencies and large positive returns of the financing currencies.

Also, the Sharpe ratio suggests that the improvement of the statistics of the long leg tends to profit more from the signal based on the average forward discount than from the signal based on volatility. Indeed, the Sharpe ratio of the long leg increases from 0.62 to 0.83 (p-v 0.04) when we time the investment with the signal based on forward discounts. Inversely, accounting only for volatility a la Moreira and Muir (2017) would produce a mild change of the Sharpe ratio of the long leg from 0.62 to 0.67 (p-v 0.44). Finally, we can see that the Sharpe ratio of the short leg moves from 0.09 to 0.14 whether we use the signal based on expected returns or the signal based on market variance.

Table II about here

Similarly, when we break the signal down into the average forward premium on one side (financing currencies) and the average forward discount on the other side, we find that both sides are informative about the future return to the carry trade (see Table VII in the online Appendix A2). For instance, when we use only the average forward discount, the Sharpe ratio of the carry trade strategy jumps from 0.76 to 1.03, while with the average forward premium, it jumps to 0.92. Finally, if we aggregate both signals without balancing them[11], we obtain a lower Sharpe ratio of 1.00. Indeed, this way of building the indicator lowers the impact of the financing currencies, while those currencies convey important information about the future return of the strategy.

3.3 Leverage and practical implementation

One important question arises here: can the average investors implement our strategy? Especially, what is the maximum leverage generated by the signal? In our sample, the ratio between the maximum value of the signal and the minimum value is around 11. While this leverage is lower than the maximum leverage studied by Darvas (2009), i.e. 25, it might not be acceptable for many risk-averse investors, especially under VaR constraints. Indeed, increasing leverage also increases volatility risk, which could then lower the utility gains of the average investor. Following DeMiguel et al. (2009) and Cederburg, O'Doherty, Wang, and Yan (2020), we quantify the gains from investing in the managed strategy rather than in the benchmark by comparing their certainty equivalent return (CER). Especially, we calculate the change in CER when investors switch from the benchmark strategy to the augmented strategy as:

$$\Delta CER = \left(\mu_{aug} - \frac{\gamma}{2} Var_{aug}\right) - \left(\mu_{bench} - \frac{\gamma}{2} Var_{bench}\right) \tag{9}$$

with μ_{aug} and μ_{bench} the mean returns of the augmented and benchmark strategies and Var_{aug} and Var_{bench} their variances. For risk aversion of $\gamma = 5$, the gain is as great as $\Delta CER = 3.02\%$ per year. However, this gain is not significant (p-v 0.17); such a risk-averse investor would not favor the managed version of the strategy. In our sample, the augmented strategy produces statistically significant gains when the leverage is limited to a maximum of 1.35 (i.e. a maximum amount of 2.7 dollars). In this case, $\Delta CER = 2.56\%$ and the gain of utility is significant (p-v 0.05) for the investor. The Sharpe ratio of this constrained strategy is 1.03, and the difference with the benchmark strategy is significant (p-v 0.03). Interestingly, an investor who would use the signal to limit the exposure of the benchmark strategy in bad times only (no leverage) would still enjoy utility gains. Indeed, in this case, the ΔCER is 1.97% (p-v 0.01) and the Sharpe ratio is 0.97 (p-v

0.02). These results indicate that even low-leveraged investors can profit from the managed carry trade.

For most investors, the augmented strategy would be added to other portfolios, such as a well-diversified equity portfolio. Interestingly, when we combine such a portfolio -a portfolio mimicking the $S\&P500$ in our case- and the augmented strategy, the Sharpe ratio increases together with the CER and the volatility. We obtain the most appealing results for portfolios mixing about 80% of the equity index and 20% of the augmented strategy. In this case, the Sharpe ratio jumps from 0.41 for the $S\&P500$ to 0.61 (p-v 0.01) for the combining portfolio with $\Delta CER = 2.99\%$ (p-v 0.04). This portfolio implies an investment in the carry trade varying from 0.10 dollar (20%*0.5) to 0.54 (20%*2.7) and an investment of 0.8 dollar in the equity index[12].
As a matter of comparison, we report that we only obtain $\Delta CER = 1.95\%$ (p-v 0.04) when we replace the augmented strategy by the benchmark carry trade in the combining portfolio. More importantly, $\Delta CER = 0.95\%$ (p-v 0.05) when we compare the latter with the strategy combining the equity portfolio and the augmented carry trade. This result suggests again that investors might prefer the managed carry trade to the benchmark one even when the strategy is combined with well-diversified equity portfolios.

4 Additional Results and Robustness

4.1 Alternative measures of the signal

In this paper, we are not interested in identifying the best predictor of returns. Rather, our goal is to develop a simple approach that minimizes the pernicious effects of data snooping. In our sample, considering a fixed number of currencies for the signal yields similar results than considering, as in our version, only the attractive ones. For instance, the Sharpe ratio of the augmented strategy based on a signal calculated over a fixed number of currencies is 1.02 if this number is 4 (i.e., 4 investment currencies and 4 financing currencies) and 1.00 if this number is 2 (see Table VII in the online appendix A2). This is not surprising since, in our sample, those currencies with the largest discount and premium are almost always attractive. The subsets match almost perfectly. However, reliance on the notion of attractiveness offers robust theoretical ground for the definition of the signal. We also point out that the strategy differs clearly from a trend following carry trade in which the investment would be scaled by the returns or the sign of the returns observed during the preceding periods. A simple version of this strategy has a very low Sharpe ratio in our sample (see Suh, 2019, for a well-performing strategy based on past returns and forward discounts). Finally,

we report that the conclusions of this paper are unchanged when we rebalance the portfolio of carry trade positions in t using the contemporaneous signal observed in t and not the lagged signal observed in t-1, as in our benchmark augmented strategy. In this case, the Sharpe ratio of the strategy is 1.03 (p-v 0.02) against 1.07 for the benchmark augmented strategy. The main results indicate that both partial signals produce insignificant results, which stand at 0.83 (p-v 0.33) for the volatility-based signal and 0.93 (p-v 0.11) for the discount-based signal.

Concerning the denominator of the ratio, we find that using other measures of the volatility does not change the results. For instance, Barroso and Santa Clara (2015a) consider only the square of the returns instead of Equation (3). This removes the adjustment for autocorrelation from the estimation of the market variance. Using this definition, we obtain a very similar Sharpe ratio of 1.13, and when we use the variance as a measure of volatility as in Moreira and Muir (2017), we obtain 1.14. However, calculating volatility over a longer period of time such as two months instead of one lowers the Sharpe Ratio, which becomes insignificantly different from the benchmark strategy when we look backward more than 3 months. Finally, we have also tested the implied volatility as measured by the EM-VXY index as calculated by JPMorgan and available in Datastream (See the online Appendix A1b). In this case, we obtain a lower Sharpe ratio (0.87), but it is still significantly different from the benchmark strategy (p-v 0.02).

4.2 Extreme events

In Figure 4, we report the 36-month rolling Sharpe ratios (4a) of both the standard carry trade and the managed version together with their 12-month rolling Sharpe ratios (4b). We see that the managed version almost continuously exhibits a higher risk-adjusted return than the benchmark strategy in both graphs. This shows that the improvement of the statistics does not hinge on some special events or a special period of time. Interestingly, while the managed carry trade tends to suffer from the market rebound of 2003 (i.e. the end of the sell-off related to the dot-com bubble), it rather profits from the market rebound of 2009 (i.e. the end of the sell-off related to the subprime crisis). Also, since the crisis of 2008, while the benchmark carry trade exhibits a Sharpe ratio of 0.06, the Sharpe ratio of the augmented version is 0.16; the higher Sharpe ratio of the managed carry trade does not particularly hinge on the observations made for the beginning of the period. We would reach largely similar conclusions for a strategy managed using only the volatility indicator (1/V as in Figure 4c) or using only the forward discount indicator (E as in Figure 4d). Both indicators convey important information.

Table III reports the descriptive statistics of the strategy during both bad and good times. We define bad times as the periods of time between the peak and trough of the S&P500 during the burst of the dot-com bubble (08-2000 to 03-2003) and the subprime crises (09-2007 to 03-2009). Other periods of time are considered good times. The results in Table III show that the strategy works both in bad times and in good times on average. Also, we see that the signals based on volatility and forward discount both improve the strategy in bad times and in good times. However, only when they are combined in the forward-looking signal do they significantly increase the Sharpe ratio of the strategy (p-value of 0.01 in good times and 0.03 in bad times).

Table III about here

We controlled for idiosyncratic risk by successively removing each currency from the sample, one by one. We obtain largely similar results in each sample; our conclusions do not depend on a particular currency. Finally, we have controlled for outliers by winsorizing the signal at the 1% and 5% levels without noticing significant changes in the results. For instance, when we winsorize the excess returns at the 1% level, the augmented strategy exhibits a Sharpe ratio of 1.04 up from 0.72 for the benchmark strategy. The difference in Sharpe ratio is still significant (p-value of 0.01). Similarly, at the 5%, the Sharpe ratio of the augmented strategy is 1.11 up from 0.80 (p-value of 0.01).

4.3 Alternative portfolios

Our next test concerns the importance of the way in which we build the carry trade strategy. In Tables IV and V, we report the statistics for other forms of the carry trade strategy. We specifically consider $Carry_n$ when $n = 3$ and $n = 5$, $Carry_{all}$ for which we buy all the currencies for which $F/S > 1$, and we sell all the currencies for which $F/S < 1$. Also, we consider a version of $Carry_4$ in which we weight the currencies by their own discount (e.g., Asness et al., 2013 and Appendix A). This emphasizes extreme currencies, and we call this version $Carry_{size}$. Finally, we report the results for a strategy in which we consider only attractive currencies as in Barroso and Santa Clara (2015b). We call it $Carry_{ba}$ (see Appendix A for a precise definition of these strategies). For every strategy, we report the statistics for both the benchmark and augmented versions. For instance, $Carry_3$ is the benchmark strategy when $n = 3$, while $Carry_{3a}$ is the augmented version when $n = 3$. The Sharpe ratios of all strategies increase significantly when we time the mean variance

trade off with the signal: the improvement we observe is not dependent on a particular way to build the portfolio of carry trade positions.

Table IV and V about here

4.4 Other samples

Now, we perform the tests of the preceding sections again in three other samples. First, we include data for the period from November 1983 to December 1998 for currencies from the following countries: Australia, Belgium, Canada, Czech Republic, Denmark, France, Germany, Hong Kong, Hungary, India, Italy, Japan, Kuwait, New Zealand, Mexico, the Netherlands, Norway, the Philippines, Saudi Arabia, Singapore, South Africa, Sweden, Switzerland, Taiwan, Thailand, United Arab Emirates, and the UK. This dataset increases the time series available. However, as already mentioned, this sample has several features that limit the conclusions one can draw. Specifically, the dataset includes the rocky 1990s (e.g. Bekaert and Hodrick, 2018), which was a period of unstable country risk (Mexico, Asian countries) characterised by CIP violation for certain currencies, as mentioned in Lustig et al. (2011). Also, before 1997, observations for emerging markets are scarce (we have only data for Hong Kong, Singapore, and South Africa), which limits the possibility for diversification. Finally, the dataset comprises several pegged currencies (Hong Kong and Singapore) and quasi-pegged converging currencies (Belgium, France, Italy, and the Netherlands), which account for up to 35% of the sample before 1997.

In Table VI, we report the summary statistics of the carry trade benchmark strategy and the augmented version in this larger sample. The Sharpe ratio of the benchmark strategy is 0.71, while the Sharpe ratio of the augmented carry trade reaches 0.91. The difference is significant with a p-value of 0.02, which confirms our preceding conclusions. Also, neither the signal based on volatility (p-v 0.09) nor the signal based on the forward discount (p-v 0.72) produce a significant improvement of the benchmark strategy. Only their combination improves the statistics, especially by a reduction of the tail risks as measured by the skewness and the ratios to mean drawdown and maximum drawdown. In this sample, the exposure varies from 0.23 to 1.66, implying an investment varying from 0.46 dollar to 3.3 dollars. As in the main sample, the representative investor would limit the leverage at 2.7 dollars to obtain $\Delta CER = 1.73\%$ (p-v 0.05). The Sharpe ratio of this constrained strategy is 0.90, and the difference with the benchmark strategy is significant

(p-v 0.01). We report more results in this sample in the online appendix. They do not change our conclusions. We only mention that the improvement tends to come first from the short leg of the strategy in this sample against the long leg in our main sample. This confirms that our signal conveys important information for both sides of the strategy. In the online appendix, we also further extend our sample to March 1976 using data available in Datastream for some European countries and the US. Again, the tests we perform do not change our conclusions.

Table VI about here

Finally, we look at the behavior of our indicator in a smaller sample reduced to the G10 currencies for which we have data since November 1983. The countries in this sample are Australia, Belgium, Canada, Denmark, Euro Area (Germany before 1999), France, Italy, Japan, Netherland, New Zealand, Norway, Sweden, Switzerland, and the UK. We cleaned this data set following Darvas (2009).

In this sample, the improvement is still visible, but the strategy fails to produce a difference in the Sharpe ratios that would be statistically significant (p-value of 0.75 in Table XIV in the online appendix A5); the strategy does not work for G10 currencies. The results are comparable to those of the period from January 1999 to July 2018 (p-value of 0.50 in Table XV in the online appendix). Cenedese et al. (2014); Daniel, Hodrick, and Lu (2017); and Maurer, To, and Tran (2020) also report weaker results for G10 currencies. Only Maurer, Pezzo, and Taylor (2019) report significant results in this setup. We also tested a version of the strategy in which we include all currencies in $Signal_t$ (i.e. 34 currencies from G10 and EM countries). Indeed, Habib and Stracca (2012) suggest that G10 currencies returns might depend on global risk aversion, especially when they act as a safe haven for investors. Interestingly, in this case, we observe a significant jump in the Sharpe ratio from 0.44 to 0.63 (p-value 0.05 in Table VII). Also, the skewness of the augmented strategy is positive, which is in line with the results obtained earlier and compatible with the findings of Bakshi and Panayotov (2013) for the G10 currencies.

Table VII about here

To check that our conclusion does not hinge on the choice of a certain numeraire, we test our

indicator when the currencies are quoted against the EUR. The results we obtain are largely similar. For instance, in our benchmark sample of 34 currencies, the Sharpe Ratio of the basic strategy is 0.74 for a European investor against 0.76 for a US investor. The augmented strategy exhibits a Sharpe Ratio of 1.05 (p-value 0.02) against 1.07 for the US investor (not reported).

5 Complementary Tests

In this section, we account for higher-order moments of the distribution and notably the skewness for portfolio choices. The standard mean-variance framework ignores this characteristic. When investors' utility functions remain unknown, one can rely on stochastic dominance to set straightforward rules of choices. In particular, we are looking for a strategy that would be preferred by an investor who is insatiable and risk averse and who prefers positive skewness (e.g. Arrow, 1971). This strategy would be third-order stochastic dominant, meaning that it would dominate any other one for the class of utility functions satisfying $u' > 0$, $u'' < 0$ and $u''' > 0$. We compare the properties of the benchmark carry trade and the augmented version by estimating the difference of their cumulative empirical distribution functions (first-order dominance), the integrated difference of their cumulative empirical distribution functions (second-order dominance), and the double integral of this difference (third-order dominance), as in Vinod (2004).

Table VIII reports the results of the stochastic dominance tests. The cumulative sums are used as the overall indicator of dominance. Negative numbers indicate a rejection of the hypothesis that the benchmark strategy dominates the augmented one. Table VIII indicates that numbers for second- and third-order dominance are all negative in both panels. An insatiable, risk-averse investor who prefers positive skewness would not prefer the benchmark strategy to the augmented one. This makes sense, since we find that the augmented strategy tends to exhibit smaller negative skewness than the benchmark strategy. We even obtain a positive skewness in certain samples. However, we notice that well-diversified portfolios such as $Carry_{ba}$ might be preferred in their benchmark version whose first order dominates the augmented one.

Table VIII about here

Finally, we would like to compare our augmented carry trade strategy to an optimally weighted portfolio. As mentioned in the introduction of this paper, for many years, the literature on portfolio construction has compared naive strategies to optimized ones. Recently, Daniel, Hodrick, and

Lu (2017) report that optimizing produces worse results than volatility timing for the carry trade, while Maurer, To, and Tran (2020) report larger Sharpe ratios for optimized carry trade (before transaction costs).

In line with our main portfolio setup, $Carry_4$, we construct an optimized portfolio including the four lower yielding currencies and the four higher yielding currencies. We consider a version of the strategy in which the sum of the weights of the currencies must be zero, but the sum of their absolute value is unconstrained, which allows for varying leverage in the time series (varying notional value). Our approach picks optimal weights to define the portfolio with the (ex-ante) maximum Sharpe ratio. We use the forward discount and premia of the currencies as expected returns. We use a rolling window to estimate the covariance matrix and the optimal weights. We test several sizes of windows from $n = 36$ months to $n = 96$ months of observations. The Sharpe ratios are calculated in the same sample for the optimized, augmented, and benchmark strategies. To limit the well-known instability of the optimization process, we rely on the shrinkage estimation of the matrix of Ledoit and Wolf (2004). The shrinkage target is given by a one-factor model, which takes the Lustig et al. (2011) DOL as a factor[13] as in Maurer, To, and Tran (2020).

Table IX reports the Sharpe ratio of $Carry_4$ for the optimized version of the strategy and for the augmented version $Carry_{4a}$ in the sample of 34 currencies from 1999 to 2018 and from 1983 to 2018 for 39 currencies. Table IX reports the results for a comparable level of leverage. When we use a rolling window of $n = 36$, the unconstrained optimized carry trade exhibits a Sharpe ratio of 0.98 against 0.72 for the benchmark $Carry_4$. However, this difference is not significant (p-value 0.11). Maurer, To, and Tran (2020) report similar results when they account for transaction costs[14] in their sample of 29 currencies.

Now, when we look at the managed version of the carry trade, we see that the Sharpe ratio increases in this sample to 1.08, which is significantly different from the benchmark strategy (p-value 0.01). The results yield similar conclusions when we increase the estimation period n from 36 to 96 months. They all indicate that the augmented version of the strategy does better than both the benchmark and the optimized ones. However, it is important to note that the difference between the optimized version and the augmented one is insignificant. In other words, we can obtain comparable results by timing the market with our indicator or with the covariance matrix, as in Maurer et al. (2020). This suggests that the observed overperformance of the optimized carry trade compared to the benchmark one tends to come primarily from market timing rather than from portfolio construction. We can reach the same conclusions in the larger sample starting in

1983, even though, in this case, the optimized version exhibits a significantly higher Sharpe ratio than the benchmark version (but, again, not significantly different from the augmented version). As mentioned previously, our objective is not to rank the alternative methodologies. Each greatly contributes to our understanding of the foreign exchange market. Yet, a clear advantage of our signal-based methodology is that it can be reproduced and implemented easily by investors.

Table IX about here

6 Why is there an improvement?

Obviously, the improvement we observe comes from the ability of the indicator to discriminate, ex ante, between periods during which the Sharpe ratio of the benchmark strategy is high and periods during which it is low. This is visible in the data; for instance, for the 20% smaller values of the indicator, the Sharpe ratio of the benchmark strategy is -0.20, while for the 20% larger values, the Sharpe ratio is 1.14 (not reported). When investors decide on the amount invested in the strategy on the back of the signal, they simply control for the expected return to risk of the strategy. To better understand the contribution of the signal to the augmented strategy, in this section we relate the average returns to the augmented carry trade to the Uncovered Interest rate Parity (UIP) and to traditional risk factor models.

6.1 The typical Fama regression is regime dependent

In theory, the beta to the well-known Fama regression should be equal to one. However, empirical evidence shows that the coefficient varies over time, meaning that UIP does not hold in some periods but does hold in others (see Bekaert and Hodrick, 2018). Our observations suggest that the augmented carry trade exploits this instability. Notably, Figure 5a reports the five-year rolling beta of the regression of the cross-sectional monthly average change in spot value for the time-varying four higher-yielding currencies on their cross-sectional average forward discount. This is the Fama regression estimated on a rolling window, as in Bekaert and Hodrick (2018). According to the UIP, this beta should not be significantly different from 1. In this case, the forward rate would be an unbiased forecast of the future spot exchange rates. However, we can see dramatic instability in the coefficient of the regression, which varies in a range of around $[-5; +4]$ in our sample. In particular, we see long periods of time during which the beta is negative, indicating that high-yielding currencies tend to appreciate on average rather than depreciate.

In Figure 5a, we also plot the rolling five-year average value of our signal. For ease of comparison, we set the standard-deviation of the signal equal to the standard deviation of the beta. We see that higher values of the signal (high discount forward-low volatility) commanding higher leverage are associated with lower values of the beta and inversely lower values of the signal (low discount forward - high volatility) commanding lower leverage are associated with larger values of the beta. This observation suggests that our signal tends to discriminate between the up and down regimes of the beta.

This echoes the results of Clarida, Davis, and Pedersen (2009), who report that the coefficient to the Fama regression might be regime dependent. They especially suggest that, when the FX market volatility is low, the carry trade is profitable, and the coefficient to the regression is negative and significant. Inversely, when FX volatility is high, the carry trade is unprofitable, and the coefficient is greater than one and significant. Using our signal, we can extend this conclusion to the value of the forward discount attractive to investors. Indeed, in Figure 5b, we see that, when we group five-year rolling betas by the value of the forward discount (i.e. the numerator of the signal $E(\pi_{t+1})$), larger betas are associated with lower values of the forward discount and vice versa. This observation suggests that the coefficient to the Fama regression might also depend on the conditional value of the forward discount attractive to investors. The UIP tends to fail (negative beta) when volatility is low and the forward discount attractive to investors is large.

Finally, in Figure 5c, we group the mean excess return of the augmented strategy over the benchmark one by the value of the beta of the Fama regression. We can see that, on average, the augmented carry trade tends to overperform the benchmark version for any value of the beta (i.e. discount/volatility regime). Interestingly, the overperformance might be greater when the beta is small to negative, i.e. when the carry trade is profitable and smaller, albeit still positive when the beta is large and positive (unprofitable carry trade). As a confirmation, we see in Figure 5d that the mean excess return of the augmented strategy over the benchmark strategy tends to be greater when the mean signal is higher (i.e. when the beta is lower). This would indicate that, using rolling mean observations, our signal is particularly useful to control leverage when the carry trade has been profitable and the beta is negative. However, following Hassan and Mano (2019), we note that the beta to the regression might not be informative about the actual and future return to the carry trade. Rather, the beta captures realized elasticity of the foreign exchange to the forward discount over a long sample (five years in our case). Only the contemporaneous discount/volatility regime seems to give indications about the future return to the carry trade and the relationship between changes in spot exchange rates and forward discounts.

6.2 Augmented carry trade exposures to risk factors

The standard Asset Pricing Theory justifies the return to the carry trade through compensation for risk. In this section, we follow Daniel, Hodrick, and Lu (2017) and Bekaert and Panayotov (2020), who run regressions of carry trade returns on returns of assets or portfolios F_t that represent the sources of risk. We especially want to know whether the augmented carry trade exhibits different exposures to popular risk factors than the benchmark carry trade. Daniel, Hodrick, and Lu (2017)

and Bekaert and Panayotov (2020) conclude that risk factor models justify the carry trade returns very little, which partially remains a puzzle.

We examine the time series regarding whether a linear combination of factors can significantly justify the returns to the carry trades. This regression is similar to the first step of the procedure inspired by Fama and MacBeth (1973):

$$R_{t+1} = \alpha_i + f'_{t+1}\beta_i + \epsilon_{it+1} \tag{10}$$

We test several factor models. The results are reported in Table X for the period 1999 to 2018. In this sample, the average return of the augmented carry trade we try to justify is 0.7% (for a monthly standard-deviation of 2.3%). The results for the period from 1983 to 2018 are reported in the online appendix. The conclusions are similar.

First, we present the results of regressions of R_{it+1} on the three Fama-French (1993) and the more recent five Fama-French (2015) equity market risk factors. The data are from French's web site, and the sample is from January 1999 to July 2018. The factors are the equity excess market return, the return on a portfolio of small-market capitalization stocks minus the return on a portfolio of big stocks, and the return on a portfolio of high book-to-market stocks minus the return on a portfolio of low book-to-market stocks for the three-factor model. Fama and French (2015) add the return on a portfolio of robust operating profitability minus the return on a portfolio of weak operating profitability and the return on a portfolio of conservative investment minus the return on a portfolio of aggressive investment to build their five-factor model. Looking at Table X, we see that the equity market factors explain little of the average return to the augmented carry trade. Indeed, in columns 1 and 2, the constant terms are large and significant at around 0.6% while, among the factors considered, only the market return is significant but not economically meaningful. Also, the R^2 is only 0.07. This is in line with the results obtained by Bekaert and Panayotov (2020) and Daniel, Hodrick, and Lu (2017) in a smaller sample of G10 currencies. The return to the augmented carry trade cannot be justified by exposure to the equity market, even when we include emerging market currencies in the sample.

To capture the possible exposure to volatility, we follow Daniel, Hodrick, and Lu (2017) and introduce the return on a variance swap as a risk factor. This return is calculated as:

$$R_{VS,t+1} = \sum_{d=1}^{Ndays} \left(ln \frac{P_{t+1,d}}{P_{t+1,d-1}} \right)^2 \left(\frac{30}{Ndays} \right) - VIX_t^2 \tag{11}$$

where $Ndays$ is the number of trading days in a month, and $P_{t+1,d}$ is the value of the $S\&P$ 500 index on day d of month $t+1$. When we add $R_{VS,t+1}$ to the regressions, we see that the results are mixed; for the case of the three-factor model, all of the loadings and the constant term are insignificant, but in the case of the five-factor model, the volatility indicator together with the market return are significant. The latter observation indicates that the carry trade does poorly when equity volatility increases. Also, in the case of the five-factor model, the constant term becomes insignificant, which might indicate that the return to the carry trade is largely justified by the exposure of the strategy to the equity market risk. However, the constant term remains economically meaningful, which also indicates that these factors do not capture the full story[15]

Now, in columns 5 and 6 of Table X, we test pure FX factors as proposed by Lustig et al. (2011) and Asness et al. (2013). The factors are estimated in our sample. RX is the cross-sectional average return of all currencies in the sample. HML-FX is the return of the benchmark carry trade when the currencies are sorted in six portfolios based on their forward discount relative to the dollar. Finally, MoM is the return of a momentum strategy when the currencies are sorted into six portfolios based on their past 11-month return with a lag of one month. By construction, HML-FX captures a large part of the return to the augmented carry trade. As a result, the R^2 of the regression is much higher at 0.56. However, the unexplained part of the augmented carry trade remains significant at the 5% level. When we add the momentum factor to the regression (column 6), we see that it significantly justifies a share of the carry trade return. However, very little changes within the constant term, which remains highly significant.

Finally, we test bond market risks. Columns 7 and 8 of Table X present the results of regressions of the augmented carry trade returns on the excess equity market return and the excess return on the 10-year bond over the one-month bill rate, which represents an interest rate level factor and the difference in returns between the 10-year bond and the two-year note, which represents a slope factor. The bond market returns are extracted from Datastream. Again, the loadings of these factors are non-significant, and the largest share of the return of the augmented carry trade ends up in the constant term. Also, when we include the volatility factor $R_{VS,t+1}$ in the regression, the results are largely similar. The loading is negative and significant, but the exposure is not significant enough to explain the profitability of the carry trade, as the constant term remains economically meaningful.

We conclude that, together with the carry trade, the augmented carry trade is poorly explained by conventional factors. Only the equity market return and volatility tend to justify it, even though

not in a significantly meaningful way[16]. Also, we find that the augmented carry trade is not priced by the benchmark carry trade HML-FX. Managing the risk exposure of the strategy based on the signal $\frac{E(\pi_{t+1})}{(MV_t)}$ produces a significant and large alpha.

Table X about here

7 Conclusion

In this article, we condition the amount invested in the carry trade by a measure of the expected global risk-adjusted return of the strategy. Our work follows Cenedese et al. (2014); Barroso and Santa Clara (2015a); Ackermann, Pohl, and Schmedders (2016); Daniel and Moskowitz (2016); Daniel, Hodrick, and Lu (2017); Moreira and Muir (2017); Maurer, Pezzo, and Taylor (2019); and Maurer, To, and Tran (2020). This study differs from the existing literature, because our signal is continuous and not binary, in contrast to Cenedese et al. (2014). Also, our indicator is based on a time-varying proxy of the expected returns of the strategy and not on constant expected returns, as in Moreira and Muir (2017). Finally, we do not forecast returns as in Daniel and Moskowitz (2016), as we rely only on past observations, and we do not rely on optimised portfolios (e.g. Ackermann, Pohl, and Schmedders, 2016; Maurer, To, and Tran, 2020). Our main contribution is as follows: we show that the partial signal based on forward discounts conveys as much information as volatility timing a la Moreira and Muir (2017). In addition, the combination of both signals based on the forward discounts and on volatility in a comprehensive indicator further improves the statistics of the carry trade. Specifically, the Sharpe ratio of our risk-adjusted managed strategy is significantly larger than that managed on volatility timing only. As a result, investors should favor this (E/V) indicator over simple (E) or (1/V) signals. Together, we show that investors might obtain comparable risk-adjusted returns by investing in optimised carry trade or in our augmented strategy, indicating that market timing instead of portfolio construction plays a key role.

We approximate the global expected return by the total discount to be available and attractive for investment in the market. This measure is standardized by the number of currencies available for investment, especially those for which the cost of trading (i.e. the bid-ask spread) does not outweigh the discount. Together, the expected risk is approximated by the observed volatility of the currency market during the preceding month. An investor who would take more risk when the risk return trade-off is attractive and less risk during other occasions would earn superior returns; the Sharpe ratio of such a managed strategy would reach 1.07 compared to 0.76 for the benchmark strategy in a sample of 34 currencies over the period 1999 to 2018.

Our strategy profits first of all from a reduction in the downside risk, as measured by the skewness of the distribution returns. While the benchmark strategy exhibits a negative skewness (-0.76), this statistic becomes positive for the managed strategy (0.97). Finally, we show that the performance of the augmented carry trade cannot be justified by popular factor models extracted from the equity market, the bond market, or the FX market. This finding indicates that the return

to the carry trade remains a puzzle, at least partially (e.g. Jorda and Taylor, 2012; Barroso and Santa-Clara, 2015b; Daniel et al., 2017; Sokolovski, 2019 and Bekaert and Panayotov, 2020, among others)

Our results are robust to various measures of the signal and independent from the way we build the portfolios. The strategy works in different samples, including a total of 39 currencies over the period 1976 to 2018, for which the Sharpe ratio increases from 0.77 to 0.91. The most recent observations available in the sample show that the signal is still useful, even when the carry trade exhibits very low risk-adjusted returns. Also, our results suggest that the benchmark strategy would not be preferred to the augmented strategy by an investor who would be insatiable and risk averse and who prefers positive skewness, even when subject to hard leverage constraints. In this paper, our objective is not to dismiss or outperform the alternative methodologies to time the carry trade that may exist. Each contributes greatly to our understanding of the foreign exchange market. However, a clear advantage of our signal-based methodology is that it can be reproduced and implemented easily by investors.

Appendix A: Portfolio design.

In this paper, we study the benchmark equally weighted carry trade as described in Equation (6), Equation (7), and Equation (8). We refer to this as $\mathbf{Carry_n}$. For robustness tests, we also study alternative portfolio constructions as presented below:

$\mathbf{Carry_{size}}$ In this strategy, the currencies with a more significant discount or premia weigh more in the return to the strategy (e.g, Asness et al., 2013 and Daniel, Hodrick and Lu, 2017, among others). The portfolios of carry trade are:

$$\begin{cases} Buy & \mathrm{w}_i = f_t^i/s_t^i - 1 \quad \text{for the n currencies i with the higher yields,} \\ Sell & \mathrm{w}_j = f_t^j/s_t^j - 1 \quad \text{for the n currencies j with the lower yields,} \\ 0 \quad otherwise & \end{cases} \tag{12}$$

The return of the portfolios are:

$$\begin{cases} R_{t+1}^{long} = \frac{1}{(\sum w_i)} \sum_{i=1}^{n} w_i * r_{t+1}^i \\ R_{t+1}^{short} = \frac{1}{(\sum |w_j|)} \sum_{j=1}^{n} w_j * r_{t+1}^j \end{cases} \tag{13}$$

with $\sum w_i$ as the total weight of the investment currencies or similarly the sum of the forward discount of the investment currencies. $\sum |w_j|$ is the absolute value of the total weight of the financing currencies or similarly the absolute value of the sum of the forward premium of the financing currencies. Finally, with $w_i > 0$ and $w_j < 0$, the return of the carry trade strategy is the sum of R_t^{long} *and* R_t^{short}(see Equation (5) for a definition of r_{t+1}).

$\mathbf{Carry_{ba}}$ To apply this strategy, every month, traders buy and sell only those attractive currencies, as defined by Equation (1). Hence, the number of currencies in the portfolios is time-varying (e.g., Burnside et al., 2007). However, the strategy is self-financing, holding a long position of one dollar in the investment currencies and a short position of one dollar in the financing currencies. We call this design $Carry_{ba}$ because the time-varying transaction costs, the bid-ask spreads, define the set of investable currencies for every period.

$$\begin{cases} Buy & \mathrm{w}_i = 1 \quad \text{if} \quad f_t^b/s_t^a > 1, \\ Sell & \mathrm{w}_j = -1 \quad \text{if} \quad f_t^a/s_t^b < 1, \\ 0 \quad otherwise \end{cases} \tag{14}$$

The return of the portfolios are:

$$\begin{cases} R_{t+1}^{long} = \frac{1}{(nl)} \sum_{i=1}^{nl} w_i * r_{t+1}^i \\ R_{t+1}^{short} = \frac{1}{(ns)} \sum_{j=1}^{ns} w_j * r_{t+1}^j \end{cases} \tag{15}$$

with nl as the number of currencies in which the investor has a long position and ns as the number of currencies in which it has a short position. Finally, with $w_i > 0$ and $w_j < 0$, the return of the carry trade strategy is the sum R_t^{long} *and* R_t^{short}.

Carry$_{\text{all}}$ When agents consider that transaction costs are insignificant, they do not discriminate between attractive and non-attractive currencies. In this case, they might form equally weighted portfolios containing all the currencies together. The investors' allocation rule is:

$$\begin{cases} Buy \quad \mathrm{w}_i = 1 \quad \text{if} \quad f_t/s_t > 1, \\ Sell \quad \mathrm{w}_j = -1 \quad \text{if} \quad f_t/s_t < 1, \end{cases} \tag{16}$$

The return of the portfolios are:

$$\begin{cases} R_{t+1}^{long} = \frac{1}{(nl)} \sum_{i=1}^{nl} w_i * r_{t+1}^i \\ R_{t+1}^{short} = \frac{1}{(ns)} \sum_{j=1}^{ns} w_j * r_{t+1}^j \end{cases} \tag{17}$$

with nl representing the number of currencies in which the investor has a long position and ns representing the number of currencies in which she has a short position. Finally, with $w_i > 0$ and $w_j < 0$, the return of the carry trade strategy is the sum R_t^{long} *and* R_t^{short}. We call this design

$Carry_{all}$.

Appendix B: Risk statistics

As with most financial returns, the returns to the carry trade are not normally distributed. They exhibit large negative skewness and substantial kurtosis (see Table I). Therefore, the two first moments of the distributions are not sufficient to describe them. In this Appendix, we introduce several popular risk ratios that have been developed by academics and the financial industry to account for non-normal distributions and complement the traditional Sharpe ratio. These ratios relate the mean return of the distribution to different risk statistics: the standard-deviation for the Sharpe ratio, the semi variance for the Sortino Ratio, the mean drawdown, and the maximum drawdown for cumulative downside risks measures.

We study the generic risk ratio (RR):

$$RR = \frac{mean(x_t)}{risk(x_t)} \tag{18}$$

with $risk(x_t) \in \{sd(x_t); SoR(x_t); \mathcal{D}_{mean}; \mathcal{D}_{Max}\}$

The standard deviation, sd(x), is a sufficient measure of risk only if the distribution is symmetrical. But since most distributions are not, one must focus more precisely on negative outcomes to estimate the risk. In this paper, we use the well-known Sortino ratio, which replaces the standard deviation by the square root of the mean squared distance of the negative outcomes to a target return (set at 0 in our case). We call this ratio SoR.

In addition to this measure of downside risk, we look at the risk of severe outcomes. According to Thaler and Johnson (1990), economic agents might also exhibit preferences in terms of the distributions of cumulative returns, especially negative ones, i.e. drawdowns. Long lasting episodes of negative outcomes might be particularly painful. We measure this risk by both the average and maximum drawdown observed in the series of returns. A drawdown is defined as the difference in value of the series of cumulated returns between any local maximum and the next local minimum. Obviously, less-pronounced drawdowns are preferred. With X as a vector of time series observations of length T, we define any drawdown $\mathcal{D}_t$ in X as:

$$\mathcal{D}_t = x_t^{max} - x_t \tag{19}$$

with

$$x_t^{max} = max\{x_i \,|i\,[0,t]\} \quad (20)$$

then

$$\forall x_t \quad x_t = x_t^{max} \Leftrightarrow \mathcal{D}_t = 0$$

otherwise

$$\mathcal{D}_t \geq 0$$

The mean drawdown in X is then defined as:

$$\mathcal{D}_{mean} = \frac{1}{n}\sum_{i=1}^{n} \mathcal{D}_t \quad (21)$$

n being the number of drawdowns in X.

and the maximum drawdown is defined as:

$$\mathcal{D}_{Max} = max\,(\mathcal{D}_t) \quad (22)$$

In this paper, the ratio of the mean return to the mean drawdown is called M_{DD}, while the ratio of the mean return to the maximum drawdown is called Max_{DD}.

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Notes

[1].An alternative solution to this puzzle is the peso story: risk-averse agents assign small but non-zero probabilities to rare events with larger negative payoff than can be observed in a sample. See, for instance, Barro and Ursua (2011) and Gourio (2008) for macro disasters and Burnside, Eichenbaum, Kleshchelski, and Rebelo (2011) for market disasters.

[2]Random walk: $E_t(1/s^b_{t+1}) = 1/s^b_t$ and $E_t(1/s^a_{t+1}) = 1/s^a_t$

[3]This strategy is symmetric in value and self-financing, as the typical investor would buy one dollar of the investment currencies and sell one dollar of the financing currencies. The risk of this strategy depends on both sides of the bet, hence, on 2 dollars.

[4]Turkey enters the sample in March 2004, as earlier observations contain obvious violation of the CIP.

[5]The cleaning process is conducted as follows: every time the bid and ask prices are equal for the spot exchange rate and/or the forward exchange rate, we recalculate both spot and forward ask prices by adding the previous day's bid ask spread, of the spot and forward rate to their respective current bid price. Similarly, we can recalculate both bid prices by subtracting the previous day's bid ask spreads to the current ask prices. Both methodologies yield very similar results. Our methodology is slightly different from that of Darvas (2009), who, in this case, replace the present day's observation with the previous day's observation. With our methodology, we believe we should keep the structure of the data in the time series, as we do not erase the information we have about the direction of the change in prices. Together, we do the same when the spread of the spot exchange rate is higher than the spread of the forward exchange rate. Finally, only when the spot moves but the forward stays constant and/or the forward moves but the spot stays constant, we replace the present-day observation with the previous day's price.

[6]We use log prices to calculate the returns to the portfolios while the signal in Section 2 is computed with spot and forward prices. The use of log prices for the signal or prices for the calculations of the returns does not change the conclusions to this research.

[7]Sorting on forward discounts or on interest rate is the same, as covered interest parity holds closely at daily frequencies (e.g., Akram et al., 2008)

[8]We study the payoff for the strategy of buying the currencies with a forward discount and selling the currencies with a forward premium. The payoff for this strategy differs by a factor $(1 + r^{US})$ from the strategy that borrows funds in a low-interest-rate currency and lends in a high-interest-rate currency.

[9]The number of currency n could vary on the business side from three to five currencies (Deutsche Bank, HSBC among others) and from one (e.g, Bakshi and Panayotov, 2013) to six (e.g., the HML-FX portfolio of Lustig, Roussanov and Verdelhan, 2011) in the academic literature, without any rationale for currency construction.

[10]We estimate the distribution of the statistics by generating 10000 block bootstrap samples of the carry trade returns. We present the results with non-annualized figures in order to follow Lo (2002), who shows that annualization is correct only under very special circumstances.

[11]

$$E_t(\pi_{t+1}) = \left[\sum_i (f^b_t/s^a_t)_i - 1 \mid (f^b_t/s^a_t)_i > 1 - \sum_j (f^a_t/s^b_t)_j - 1 \mid (f^a_t/s^b_t)_j < 1 \right] / \left[n^{buy}_t + n^{sell}_t \right] \tag{23}$$

[12]Sophisticated investors might also consider Fama-French factors in their portfolios. We find very similar results when we combine equity portfolio mimicking the factors returns and the augmented strategy. For instance, $\Delta CER =$ 2.95% when we invest 75% (p-v 0.04) in the SMB factor and 25% in the augmented strategy. Similarly, $\Delta CER = 3.6\%$ (p-v 0.05) for HML with 24% invested in the augmented strategy, $\Delta CER = 3.72\%$ (p-v 0.04) for RMW with 27% invested in the augmented strategy and finally $\Delta CER = 3.24\%$ (p-v 0.04) for CMA with 34% invested in the augmented strategy.

[13]We use the mean cross-sectional sample return to estimate Lustig et al. (2011) DOL factor for the shrinkage. Maurer, To, Tran (2020) use the first Principal Component as factor. We know from Lustig et al. (2011) that this

first component is precisely DOL.

[14]Table XII in the online appendix of Maurer, To, and Tran (2020) reports a Sharpe ratio of 0.55 for HML and 0.82 for their benchmark optimized strategy. According to their results, this difference is not statistically significant, with a p-value of 0.17.

[15]We have also tested an interaction term between the market return and volatility without success.

[16]We reach the same conclusion when we regress the excess return of the augmented carry trade over the benchmark one on risk factors.

Appendix C: the return to the carry trade

Table I: Portfolios' descriptive statistics

The table illustrates the mean returns, standard deviation, skewness and kurtosis of currency portfolios invested in the carry trade. $Carry_n$ is an equally weighted portfolio investing in the n currencies with the largest discount minus the n currencies with the lowest discount. We report the results for $n = 4$. The leverage of the signal augmented portfolios (aug) is controlled by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. The leverage of the discount augmented portfolios (disc) is controlled by the numerator of Equation (4) $E(\pi_{t+1})$ while the leverage of the Vol augmented portfolios (vol) is controlled by the denominator of Equation (4) $\frac{1}{(MV_t)}$. The leverage of the benchmark strategy is set at 1. The statistics are reported for annualized log excess returns accounting for transaction costs (bid-ask spreads). The data represents monthly period. We also report the annualized Sharpe Ratios (SR), the annualized Information Ratio (IR) against the benchmark strategy, the Sortino Ratio SoR, the ratios of the mean return to the mean drawdown M_{DD} and the ratios of the mean return to the maximum drawdown Max_{DD} as defined in Appendix B of this paper. Monthly ratios and their standard-errors are also reported (e.g. Lo, 2002). They are obtained by bootstrapping the sample 10000 times. The figures are given in basis points, not percentages. Column p-v reports the p-value of the test of equality of the Sharpe ratios against the benchmark strategy. Panel A uses data for 34 currencies as defined in Section 3 from January 1999 to July 2018.

Panel A: 34-CCY 99-18 - Signal augmented Portfolios											
		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
$Carry_{4_{aug}}$	*annualized*	0.1041	0.0964	0.9711	10.0525	1.0794	(0.01)	0.6649	1.3989	0.3059	0.0545
	monthly					0.3115		0.1919	0.4038	0.3058	0.0545
	sd-dev					*(0.0644)*		*(0.0536)*	*(0.1179)*	*(0.0884)*	*(0.0257)*
Panel A: 34-CCY 99-18 - Discount augmented Portfolios											
$Carry_{4_{disc}}$	*annualized*	0.0957	0.0964	-0.2778	6.3220	0.9927	(0.06)	0.6246	1.0945	0.2561	0.0490
	monthly					0.2865		0.1803	0.3159	0.2560	0.0489
	sd-dev					*(0.0712)*		*(0.0666)*	*(0.1109)*	*(0.0839)*	*(0.0204)*
Panel A: 34-CCY 99-18 - Vol augmented Portfolios											
		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
$Carry_{4_{vol}}$	*annualized*	0.0813	0.0964	-0.4912	6.2011	0.8434	(0.31)	0.2693	0.8739	0.2007	0.0314
	monthly					0.2434		0.0777	0.2522	0.2006	0.0313
	sd-dev					*(0.0723)*		*(0.0636)*	*(0.0952)*	*(0.0735)*	*(0.0169)*
Panel A: 34-CCY 99-18 - Benchmark strategy											
$Carry_4$	*annualized*	0.0738	0.0964	-0.7668	4.9553	0.7656	-	-	0.7468	0.1690	0.0277
	monthly					0.2210			0.2155	0.1689	0.0277
	sd-dev					*(0.0734)*			*(0.0834)*	*(0.0671)*	*(0.0160)*

Table II: Performance attribution

The table illustrates the mean returns, standard deviation, skewness and kurtosis of currency portfolios invested in the carry trade. $Carry_n$ is an equally weighted portfolio investing in the n currencies with the largest discount minus the n currencies with the lowest discount. We report the results for $n = 4$. The leverage of the signal augmented portfolios (aug) is controlled by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. The leverage of the discount augmented portfolios (disc) is controlled by the numerator of Equation (4) $E(\pi_{t+1})$ while the leverage of the Vol augmented portfolios (vol) is controlled by the denominator of Equation (4) $\frac{1}{(MV_t)}$. The leverage of the benchmark strategy is set at 1. The statistics are reported for annualized log excess returns accounting for transaction costs (bid-ask spreads). The data represents monthly period. We also report the annualized Sharpe Ratios (SR), the annualized Information Ratio (IR) against the benchmark strategy, the Sortino Ratio SoR, the ratios of the mean return to the mean drawdown M_{DD} and the ratios of the mean return to the maximum drawdown Max_{DD} as defined in Appendix B of this paper. Monthly ratios and their standard-errors are also reported (e.g. Lo, 2002). They are obtained by bootstrapping the sample 10000 times. The figures are given in basis points, not percentages. Column p-v reports the p-value of the test of equality of the Sharpe ratios against the benchmark strategy. Panel A uses data for 34 currencies as defined in Section 3 from January 1999 to July 2018.

Panel A: 34-CCY 99-18 - Augmented Portfolios											
		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
$Carry_{aug}^{long}$	*annualized*	0.0949	0.1089	0.8484	8.5176	0.8717	(0.03)	0.5156	0.9509	0.2058	0.0380
$Carry_{aug}^{short}$	*annualized*	0.0131	0.0653	0.0340	5.3795	0.2007	(0.30)	0.2306	0.2061	0.0428	0.0093
$Carry_{disc}^{long}$	*annualized*	0.0913	0.1089	0.1923	5.4417	0.8383	(0.04)	0.6203	0.8497	0.1826	0.0310
$Carry_{disc}^{short}$	*annualized*	0.0091	0.0653	-1.0210	13.1999	0.1399	(0.66)	0.1164	0.1349	0.0312	0.0059
$Carry_{vol}^{long}$	*annualized*	0.0730	0.1089	-0.6110	5.4799	0.6708	(0.44)	0.1741	0.6023	0.1384	0.0232
$Carry_{vol}^{short}$	*annualized*	0.0092	0.0653	-0.0792	3.0674	0.1405	(0.58)	0.1219	0.1414	0.0270	0.0065
Sample: 34-CCY 99-18 - Benchmark strategy											
$Carry_4^{long}$	*annualized*	0.0674	0.1089	-0.7882	5.2485	0.6196	–	–	0.5398	0.1219	0.0191
$Carry_4^{short}$	*annualized*	0.0064	0.0653	-0.2662	3.9619	0.0972	–	–	0.0957	0.0184	0.0053

Table III: Portfolios' descriptive statistics in good times and in bad times

The table illustrates the mean returns, standard deviation, skewness and kurtosis of currency portfolios invested in the carry trade. $Carry_n$ is an equally weighted portfolio investing in the n currencies with the largest discount minus the n currencies with the lowest discount. We report the results for $n = 4$. The leverage of the signal augmented portfolios (aug) is controlled by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. The leverage of the discount augmented portfolios (disc) is controlled by the numerator of Equation (4) $E(\pi_{t+1})$ while the leverage of the Vol augmented portfolios (vol) is controlled by the denominator of Equation (4) $\frac{1}{(MV_t)}$. The leverage of the benchmark strategy is set at 1. The statistics are reported for annualized log excess returns accounting for transaction costs (bid-ask spreads). The data represents monthly period. We also report the annualized Sharpe Ratios (SR), the annualized Information Ratio (IR) against the benchmark strategy, the Sortino Ratio SoR, the ratios of the mean return to the mean drawdown M_{DD} and the ratios of the mean return to the maximum drawdown Max_{DD} as defined in Appendix B of this paper. Monthly ratios and their standard-errors are also reported (e.g. Lo, 2002). They are obtained by bootstrapping the sample 10000 times. The figures are given in basis points, not percentages. Column p-v reports the p-value of the test of equality of the Sharpe ratios against the benchmark strategy. Panel E uses data for 34 currencies as defined in Section 3 from January 1999 to July 2000, April 2003 to August 2007 and April 2009 to July 2018 (defined as good times according to the peak and through of the S&P500). Panel F uses data for 34 currencies as defined in Section 3 from August 2000 to March 2003 and September 2007 to March 2009 (defined as bad times).

		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
Panel E: 34-CCY 99-18 - Good Times											
$Carry_{4aug}$	*annualized*	0.0944	0.0894	0.0485	3.5552	1.0560	(0.01)	0.8115	1.3043	0.2741	0.0709
$Carry_{4disc}$	*annualized*	0.0904	0.0894	-0.1075	3.9447	1.0115	(0.01)	0.8462	1.1952	0.2526	0.0637
$Carry_{4vol}$	*annualized*	0.0717	0.0894	-0.2921	4.1138	0.8026	(0.27)	0.2988	0.8646	0.1843	0.0375
$Carry_4$	*annualized*	0.0656	0.0894	-0.5502	4.4499	0.7342	–	–	0.7496	0.1605	0.0405
Panel F: 34-CCY 99-18 - Bad Times											
$Carry_{4aug}$	*annualized*	0.1398	0.1196	1.3916	9.5962	1.1685	(0.03)	0.8455	1.6100	0.3918	0.0834
$Carry_{4disc}$	*annualized*	0.1118	0.1196	-0.4967	5.7540	0.9345	(0.06)	0.5731	0.9295	0.2473	0.0650
$Carry_{4vol}$	*annualized*	0.1054	0.1196	-0.7991	7.4496	0.8808	(0.21)	0.4541	0.8446	0.2316	0.0414
$Carry_4$	*annualized*	0.0855	0.1196	-1.0778	4.8770	0.7146	–	–	0.6290	0.1604	0.0321

Table IV: Alternative portfolios' descriptive statistics

The table illustrates the mean returns, standard deviation, skewness and kurtosis of currency portfolios invested in the carry trade. $Carry_{all}$ is an equally weighted portfolio of all the currencies of the set of investment (see Appendix A). $Carry_{ba}$ is an equally weighted portfolio investing in the currencies for which the cost of trading does not outweigh the discount (ex-ante profitable currencies - see appendix A). The leverage of the signal augmented portfolios (aug) is controlled by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. The leverage of the discount augmented portfolios (disc) is controlled by the numerator of Equation (4) $E(\pi_{t+1})$ while the leverage of the Vol augmented portfolios (vol) is controlled by the denominator of Equation (4) $\frac{1}{(MV_t)}$. The leverage of the benchmark strategy is set at 1. The statistics are reported for annualized log excess returns accounting for transaction costs (bid-ask spreads). The data represents monthly period. We also report the annualized Sharpe Ratios (SR), the annualized Information Ratio (IR) against the benchmark strategy, the Sortino Ratio SoR, the ratios of the mean return to the mean drawdown M_{DD} and the ratios of the mean return to the maximum drawdown Max_{DD} as defined in Appendix B of this paper. Monthly ratios and their standard-errors are also reported (e.g. Lo, 2002). They are obtained by bootstrapping the sample 10000 times. The figures are given in basis points, not percentages. Column p-v reports the p-value of the test of equality of the Sharpe ratios against the benchmark strategy. Panel A uses data for 34 currencies as defined in Section 3 from January 1999 to July 2018.

Panel A: 34-CCY 99-18 - Benchmark and augmented Portfolios											
		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
$Carry_{allaug}$	*annualized*	0.0280	0.0409	0.5168	6.1146	0.6844	(0.03)	0.3357	0.7603	0.1782	0.0356
$Carry_{alldisc}$	*annualized*	0.0269	0.0409	0.2897	4.8254	0.6576	(0.14)	0.3503	0.7075	0.1618	0.0346
$Carry_{allvol}$	*annualized*	0.0227	0.0409	-0.4659	4.4317	0.5568	(0.69)	0.0963	0.5363	0.1259	0.0210
$Carry_{all}$	*annualized*	0.0216	0.0409	-0.4036	3.4540	0.5298	-	-	0.5059	0.1138	0.0202
$Carry_{baaug}$	*annualized*	0.0507	0.0816	0.1087	9.5864	0.6217	(0.00)	0.6380	0.6146	0.1744	0.0266
$Carry_{badisc}$	*annualized*	0.0313	0.0816	-3.0700	26.2976	0.3843	(0.60)	0.1621	0.2952	0.1084	0.0140
$Carry_{bavol}$	*annualized*	0.0379	0.0816	-0.3975	6.6040	0.4647	(0.14)	0.4034	0.4125	0.1127	0.0180
$Carry_{ba}$	*annualized*	0.0262	0.0816	-1.5405	10.6914	0.3216	-	-	0.2544	0.0764	0.0121

Table V: Alternative portfolios' descriptive statistics

The table illustrates the mean returns, standard deviation, skewness and kurtosis of currency portfolios invested in the carry trade and their augmented version. $Carry_n$ is an equally weighted portfolio investing in the n currencies with the largest discount minus the n currencies with the lowest discount. We report the results for $n = 3$ and $n = 5$. $Carry_{size}$ is a spread weighted portfolio investing in the n currencies with the largest discount minus the n currencies with the lowest discount when $n = 4$. The leverage of the signal augmented portfolios is controlled by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MVt)}$. The leverage of the discount augmented portfolios (disc) is controlled by the numerator of Equation (4) $E(\pi_{t+1})$ while the leverage of the Vol augmented portfolios (vol) is controlled by the denominator of Equation (4) $\frac{1}{(MVt)}$. The statistics are reported for annualized log excess returns accounting for transaction costs (bid-ask spreads). The data are monthly. We also report the annualized Sharpe Ratios (SR), the annualized Information Ratio (IR) against the benchmark strategy, the Sortino Ratio SoR, the ratios of the mean return to the mean drawdown M_{DD} and the ratios of the mean return to the maximum drawdown Max_{DD} as defined in Appendix B of this paper. Monthly ratios and their standard-errors are also reported (e.g. Lo, 2002). They are obtained by bootstrapping the sample 10000 times. The figures are given in basis points not percentages. Column p-v reports the p-value of the test of equality of the Sharpe ratios against the benchmark strategy. Panel A uses data for 34 currencies as defined in Section 3 from January 1999 to July 2018.

Panel A: 34-CCY 99-18 - Benchmark and augmented Portfolios											
		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
$Carry_{3aug}$	*annualized*	0.1336	0.1108	1.1969	10.3324	1.2050	(0.03)	0.6551	1.5741	0.3432	0.0695
$Carry_{3disc}$	*annualized*	0.1126	0.1108	-0.9838	12.3870	1.0163	(0.45)	0.3386	1.0008	0.2641	0.0410
$Carry_{3vol}$	*annualized*	0.1139	0.1108	-0.1498	5.1623	1.0275	(0.18)	0.4466	1.0901	0.2417	0.0415
$Carry_3$	*annualized*	0.0981	0.1108	-0.7913	5.3134	0.8849	-	-	0.8332	0.1910	0.0334
$Carry_{5aug}$	*annualized*	0.0950	0.0867	0.9934	10.0710	1.0954	(0.01)	0.6201	1.3459	0.3074	0.0573
$Carry_{5disc}$	*annualized*	0.0856	0.0867	-0.4934	7.5790	0.9866	(0.16)	0.4977	1.0036	0.2510	0.0472
$Carry_{5vol}$	*annualized*	0.0762	0.0867	-0.5095	6.2845	0.8787	(0.31)	0.2619	0.8654	0.2083	0.0340
$Carry_5$	*annualized*	0.0695	0.0867	-0.7366	4.6320	0.8016	-	-	0.7451	0.1754	0.0315
$Carry_{sizeaug}$	*annualized*	0.1620	0.1382	3.0126	17.6690	1.1722	(0.03)	0.2046	2.4338	0.6025	0.1145
$Carry_{sizedisc}$	*annualized*	0.1495	0.1382	1.1662	13.2092	1.0817	(0.06)	0.0325	1.4650	0.4665	0.0559
$Carry_{sizevol}$	*annualized*	0.1591	0.1382	1.1716	7.9976	1.1516	(0.06)	0.2650	1.6037	0.4063	0.0639
$Carry_{size}$	*annualized*	0.1477	0.1382	0.2400	5.7545	1.0689	-	-	1.2504	0.3296	0.0542

Table VI: Portfolios' descriptive statistics

The table illustrates the mean returns, standard deviation, skewness and kurtosis of currency portfolios invested in the carry trade. $Carry_n$ is an equally weighted portfolio investing in the n currencies with the largest discount minus the n currencies with the lowest discount. We report the results for $n = 4$. The leverage of the signal augmented portfolios (aug) is controlled by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. The leverage of the discount augmented portfolios (disc) is controlled by the numerator of Equation (4) $E(\pi_{t+1})$ while the leverage of the Vol augmented portfolios (vol) is controlled by the denominator of Equation (4) $\frac{1}{(MV_t)}$. The leverage of the benchmark strategy is set at 1. The statistics are reported for annualized log excess returns accounting for transaction costs (bid-ask spreads). The data represents monthly period. We also report the annualized Sharpe Ratios (SR), the annualized Information Ratio (IR) against the benchmark strategy, the Sortino Ratio SoR, the ratios of the mean return to the mean drawdown M_{DD} and the ratios of the mean return to the maximum drawdown Max_{DD} as defined in Appendix B of this paper. Monthly ratios and their standard-errors are also reported (e.g. Lo, 2002). They are obtained by bootstrapping the sample 10000 times. The figures are given in basis points, not percentages. Column p-v reports the p-value of the test of equality of the Sharpe ratios against the benchmark strategy. Panel B uses data for 39 currencies as defined in Section 3 from November 1983 to July 2018.

Panel B: 39-CCY 83-18 - Signal augmented Portfolios											
		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
$Carry_{4_{aug}}$	*annualized*	0.0839	0.0918	-0.0706	5.1258	0.9136	(0.01)	0.5086	0.9761	0.2282	0.0370
	monthly					0.2687		0.1468	0.2817	0.2281	0.0370
	sd-dev					*(0.0511)*		*(0.0466)*	*(0.0672)*	*(0.0568)*	*(0.0116)*
Panel B: 39-CCY 83-18 - Discount augmented Portfolios											
$Carry_{4_{disc}}$	*annualized*	0.0682	0.0918	-0.7454	7.3313	0.7431	(0.72)	0.0744	0.7066	0.1768	0.0206
	monthly					0.2145		0.0214	0.2039	0.1767	0.0205
	sd-dev					*(0.0546)*		*(0.0493)*	*(0.0636)*	*(0.0536)*	*(0.0124)*
Panel B: 39-CCY 83-18 - Vol augmented Portfolios											
		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
$Carry_{4_{vol}}$	*annualized*	0.0757	0.0918	-0.4740	5.7211	0.8245	(0.09)	0.3532	0.8241	0.1948	0.0293
	monthly					0.2380		0.1019	0.2378	0.1947	0.0292
	sd-dev					*(0.0536)*		*(0.0474)*	*(0.0639)*	*(0.0530)*	*(0.0104)*
Panel B: 39-CCY 83-18 - Benchmark strategy											
$Carry_4$	*annualized*	0.0655	0.0918	-0.7471	4.6275	0.7139	-	-	0.6675	0.1536	0.0246
	monthly					0.2060			0.1926	0.1535	0.0246
	sd-dev					*(0.0534)*			*(0.0555)*	*(0.0468)*	*(0.0093)*

Table VII: Portfolios' descriptive statistics

The table illustrates the mean returns, standard deviation, skewness and kurtosis of currency portfolios invested in the carry trade. $Carry_n$ is an equally weighted portfolio investing in the n currencies with the largest discount minus the n currencies with the lowest discount. We report the results for $n = 4$. The leverage of the signal augmented portfolios (aug) is controlled by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. The leverage of the discount augmented portfolios (disc) is controlled by the numerator of Equation (4) $E(\pi_{t+1})$ while the leverage of the Vol augmented portfolios (vol) is controlled by the denominator of Equation (4) $\frac{1}{(MV_t)}$. The leverage of the benchmark strategy is set at 1. The statistics are reported for annualized log excess returns accounting for transaction costs (bid-ask spreads). The data represents monthly period. We also report the annualized Sharpe Ratios (SR), the annualized Information Ratio (IR) against the benchmark strategy, the Sortino Ratio SoR, the ratios of the mean return to the mean drawdown M_{DD} and the ratios of the mean return to the maximum drawdown Max_{DD} as defined in Appendix B of this paper. Monthly ratios and their standard-errors are also reported (e.g. Lo, 2002). They are obtained by bootstrapping the sample 10000 times. The figures are given in basis points, not percentages. Column p-v reports the p-value of the test of equality of the Sharpe ratios against the benchmark strategy. Panel D uses data for 10 currencies as defined in Section 3 from January 1999 to July 2018. Panel A uses data for 34 currencies as defined in Section 3 from January 1999 to July 2018.

Panel D: 10-CCY 99-18 - Signal extracted from panel A: 34-CCY 99-18 - Signal augmented Portfolios											
		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
$Carry_{4_{aug}}$	*annualized*	0.0458	0.0720	0.1666	4.9783	0.6372	(0.05)	0.3866	0.6719	0.1370	0.0257
	monthly					0.1839		0.1116	0.1939	0.1369	0.0257
	sd-dev					*(0.0666)*		*(0.0626)*	*(0.0794)*	*(0.0578)*	*(0.0162)*
Panel D: 10-CCY 99-18 - Signal extracted from panel A: 34-CCY 99-18 - Discount augmented Portfolios											
$Carry_{4_{disc}}$	*annualized*	0.0417	0.0720	-0.2803	4.7943	0.5800	(0.15)	0.3572	0.5758	0.1198	0.0207
	monthly					0.1674		0.1031	0.1662	0.1197	0.0207
	sd-dev					*(0.0676)*		*(0.0648)*	*(0.0753)*	*(0.0545)*	*(0.0154)*
Panel D: 10-CCY 99-18 - Signal extracted from panel A: 34-CCY 99-18 - Vol augmented Portfolios											
		mean	sd	SK	KT	SR	p-v	IR	SoR	M_{DD}	Max_{DD}
$Carry_{4_{vol}}$	*annualized*	0.0364	0.0720	-0.4414	3.9446	0.5063	(0.42)	0.1979	0.4804	0.0982	0.0161
	monthly					0.1461		0.0397	0.1386	0.0982	0.0161
	sd-dev					*(0.0681)*		*(0.0659)*	*(0.0684)*	*(0.0515)*	*(0.0127)*
Panel D: 10-CCY 99-18 - Signal extracted from panel A: 34-CCY 99-18 - Benchmark strategy											
$Carry_4$	*annualized*	0.0321	0.0720	-0.4999	3.8693	0.4465	–	–	0.4122	0.0838	0.0130
	monthly					0.1289			0.1190	0.0837	0.0129
	sd-dev					*(0.0686)*			*(0.0668)*	*(0.0487)*	*(0.0124)*

Appendix D: Complementary tests

Table VIII: Skewness and portfolio choices

The table reports the results of stochastic dominance tests. The cumulative sums are used as the overall indicator of dominance, and negative numbers indicate rejection of the hypothesis that the benchmark strategy dominates the augmented one for various type of portfolios. $Carry_n$ is an equally weighted portfolio investing in the n currencies with the largest discount minus the n currencies with the lowest discount. We report the results for various values of n from $n = 3$ to $n = 5$. $Carry_{all}$ is an equally weighted portfolio of all the currencies of the set of investments (see Appendix A). $Carry_{size}$ is a spread weighted portfolio investing in the n currencies with the greatest discount minus the n currencies with the smallest discount when $n = 4$. $Carry_{ba}$ is an equally weighted portfolio investing in the currencies for which the cost of trading does not outweigh the discount (attractive currencies; see Appendix A). The statistics are reported for annualized log excess returns accounting for transaction costs (bid-ask spreads). The data represent monthly time period. Panel A uses 34 currencies from January 1999 to July 2018, and Panel B uses all available data from November 1983 to July 2018.

Panel A: 34-CCY 99-18							
	$1SD$	2SD	$3SD$		$1SD$	2SD	$3SD$
$Carry_5$	-0.0007	-0.0945	-0.6191	$Carry_{all}$	-0.0001	-0.0069	-4.6781
$Carry_4$	-0.0014	-0.1276	-7.6423	$Carry_{size}$	-0.0007	-0.0602	-4.1319
$Carry_3$	-0.0014	-0.1522	-1.0074	$Carry_{ba}$	0.0009	-0.0301	-1.4302
Panel B: 39-CCY 83-18							
	$1SD$	2SD	$3SD$		$1SD$	2SD	$3SD$
$Carry_5$	-0.0004	-0.1161	-2.2110	$Carry_{all}$	-0.0008	-0.1578	-1.1430
$Carry_4$	-0.0003	-0.1441	-2.0201	$Carry_{size}$	-0.0006	-0.0516	-1.381
$Carry_3$	-0.0004	-0.0953	-1.9950	$Carry_{ba}$	0.0007	-0.1278	-0.1461

Table IX: Augmented and optimized portfolios

The table reports the Sharpe ratios and the p-value of the tests of the difference of the Sharpe ratio for currency portfolios invested in the carry trade. $Carry_n$ is an equally weighted portfolio investing in the n currencies with the largest discount minus the n currencies with the lowest discount. We report the results for $n = 4$. The leverage of the signal augmented portfolios (aug) is controlled by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. $Carry_{OPT}$ is an optimal portfolio including the four lower-yielding currencies and the four higher-yielding currencies. The results are reported for various sizes of the covariance matrix from 36 months to 96 months. The statistics are reported for the annualized log excess returns accounting for transaction costs (bid-ask spreads). The data are monthly. Panel A uses all countries with data from January 1999 to July 2018. Panel B uses all countries with data from November 1983 to July 2018.

Panel A: 34-CCY 99-18						
Vcov size in months	36	48	60	72	84	96
SR $Carry_4$	0.72	0.58	0.44	0.30	0.13	0.11
SR $Carry_{OPT}$	0.98	0.77	0.50	0.28	0.17	0.14
SR $Carry_{4a}$	1.08	0.89	0.70	0.52	0.35	0.35
p-value $\text{Carry}_{4/OPT}$	0.10	0.23	0.73	0.91	0.81	0.83
p-value $\text{Carry}_{4/4a}$	0.01	0.01	0.01	0.02	0.04	0.03
p-value $\text{Carry}_{OPT/4a}$	0.55	0.45	0.32	0.20	0.37	0.30
Panel B: 39-CCY 83-18						
Vcov size in months	36	48	60	72	84	96
SR $Carry_4$	0.73	0.72	0.68	0.69	0.66	0.64
SR $Carry_{OPT}$	1.01	0.99	1.01	1.01	1.00	1.01
SR $Carry_{4a}$	0.91	0.91	0.86	0.87	0.87	0.86
p-value $\text{Carry}_{4/OPT}$	0.05	0.08	0.03	0.03	0.05	0.03
p-value $\text{Carry}_{4/4a}$	0.03	0.04	0.06	0.04	0.02	0.01
p-value $\text{Carry}_{OPT/4a}$	0.53	0.61	0.39	0.40	0.53	0.41

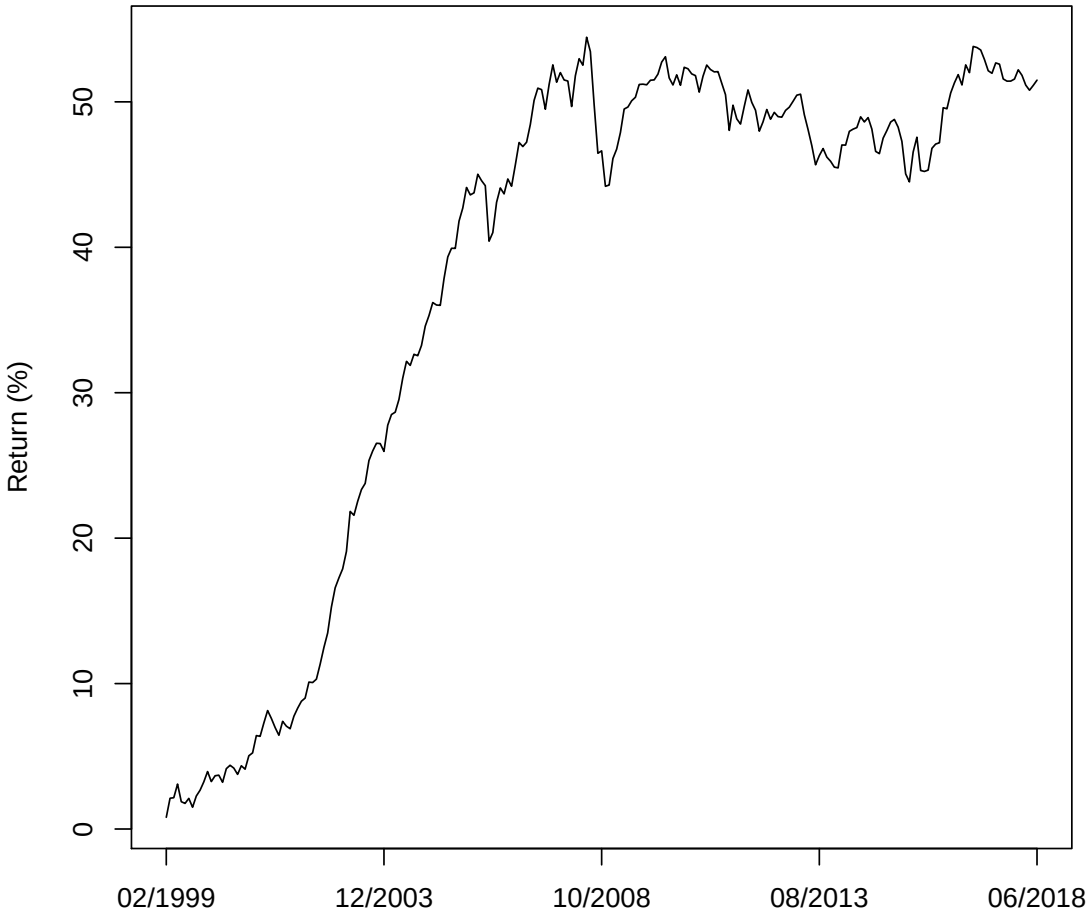
Appendix E: Asset pricing.

Table X: Augmented carry trade exposure to risk factors.

The table reports the loading of the regression of the return of the augmented carry trade strategy on factor models. FF3 and FF5 are the Fama-French three- and five-factor models. FX-HML is the model proposed by Lustig et al. (2011), and FX-HML-MoM includes the momentum strategy as proposed by Asness et al. (2013). Bond is the model proposed by Daniel et al. (2017). β_{Vol} captures the possible exposure to equity market volatility. The associated standard deviation is reported into brackets. Significant loadings at the 5% level are indicated with two*. Panel A uses data for 34 currencies, as defined in Section 3, from January 1999 to July 2018.

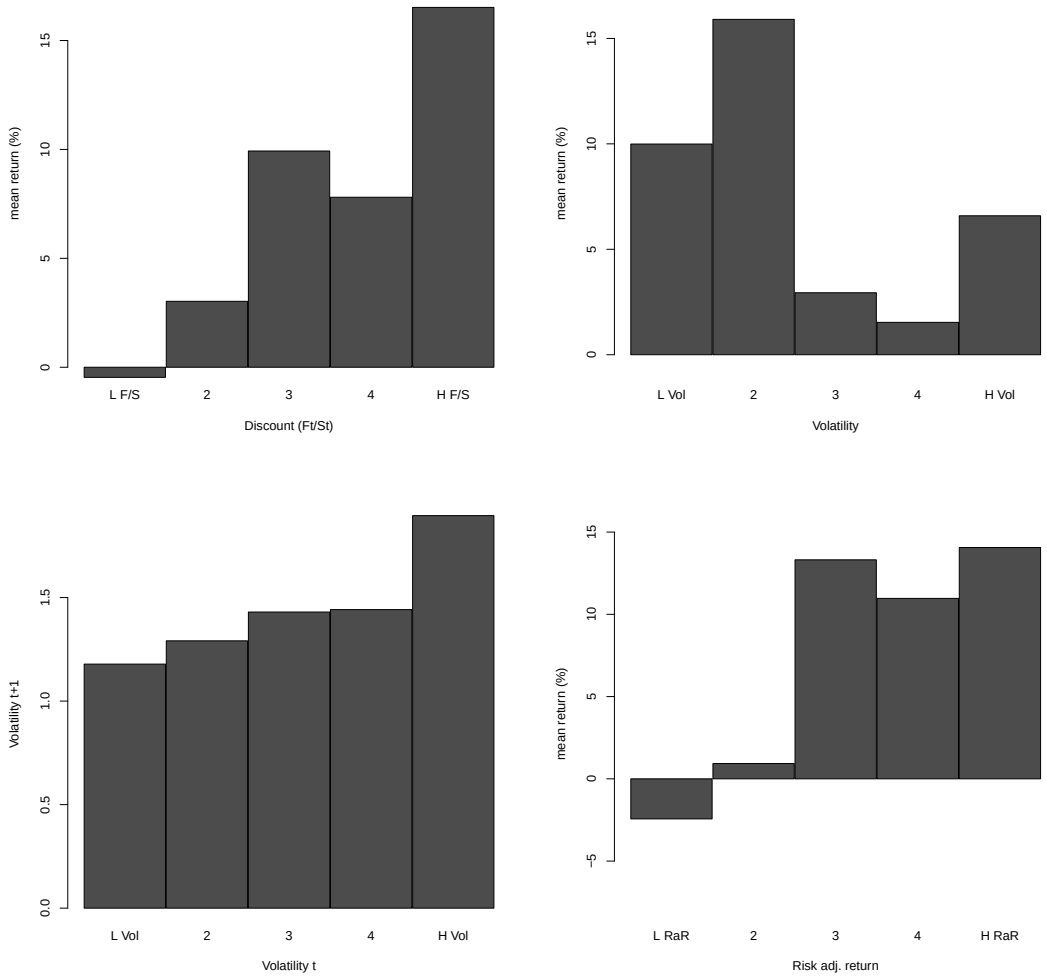
Panel A: All countries: 99-18										
	FF3	FF5	FF3-vol	FF5-vol		FX-HML	FX-HML-MoM		Bond	Bond-vol
α	0.61**	0.59**	0.28	0.19	α	0.26**	0.27**	α	0.43**	0.27
	(0.15)	(0.16)	(0.18)	(0.17)		(0.10)	(0.10)		(0.17)	(0.17)
β_{MKT}	0.12**	0.13**	0.07	0.19**	β_{RX}	-1.72	-1.08	β_{MKT}	0.21**	0.16**
	(0.04)	(0.04)	(0.05)	(0.05)		(6.10)	(5.93)		(0.04)	(0.04)
β_{SMB}	0.06	0.09	0.11	-0.01	β_{HML-FX}	0.81**	0.80**	β_{10y}	-0.81	-0.65
	(0.04)	(0.05)	(0.08)	(0.08)		(0.04)	(0.05)		(0.58)	(0.58)
β_{HML}	0.03	0.03	0.03	-0.02	β_{MoM}		4.85**	β_{10y-2y}	0.78	0.60
	(0.04)	(0.06)	(0.07)	(0.08)			(1.72)		(0.65)	(0.65)
β_{RMW}		0.04		0.08						
		(0.07)		(0.13)						
β_{CMA}		-0.05		-0.05						
		(0.09)		(0.14)						
β_{Vol}			-0.22	-0.85**				β_{Vol}		-0.82**
			(0.31)	(0.29)						(0.28)
R^2	0.07	0.07	0.05	0.09	R^2	0.56	0.60	R^2	0.14	0.20

Figure 1: The return to the carry trade



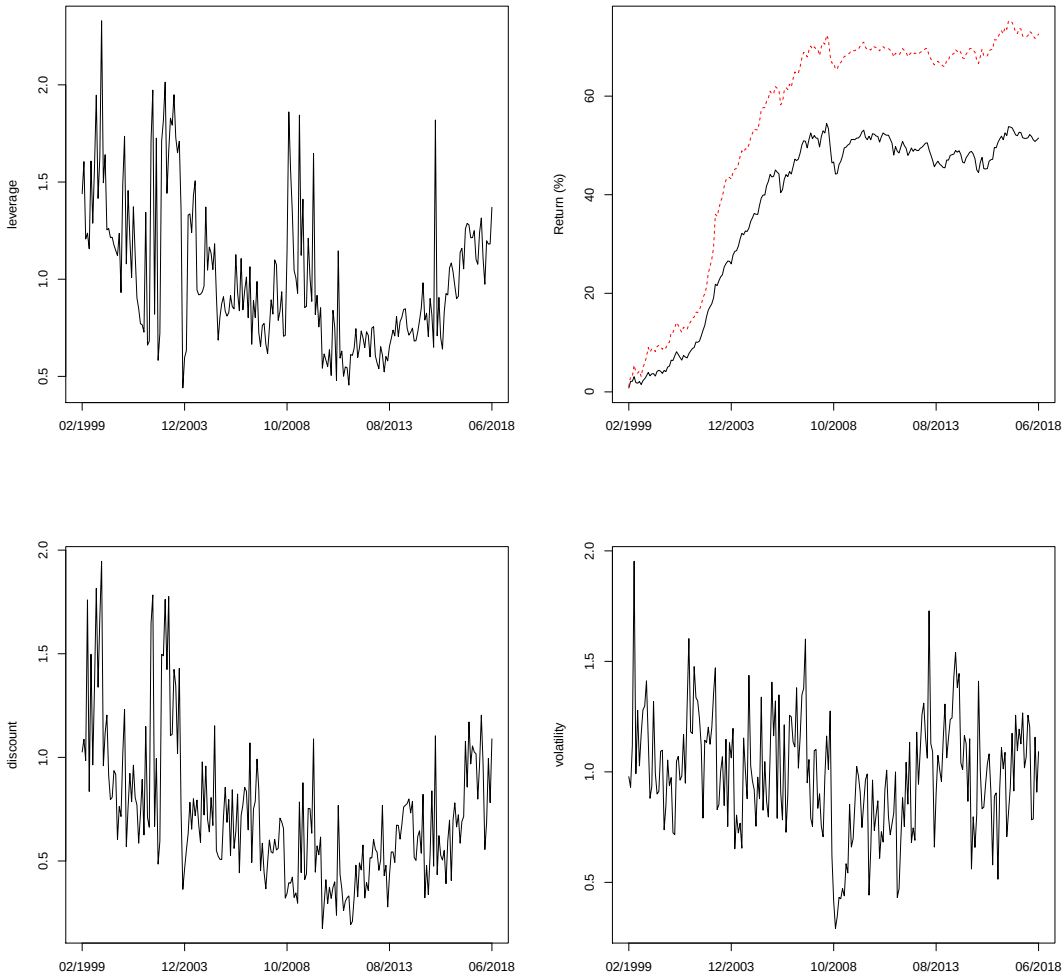
Sample from January 1999 to July 2018 for 34 currencies as defined in Section 3. Monthly observations. The figure plots the cumulated return of investing one dollar in a portfolio of long positions in currencies with high interest rates and one dollar in a portfolio of short positions in currencies with low interest rates. The strategy is rebalanced every month, and the proceeds are not reinvested. The portfolio is equally weighted and contains four currencies of each type. The strategy offers a positive mean return from 1999 to 2008, but the recent picture is more mixed. The Sharpe ratio of the strategy over the sample is 0.76.

Figure 2: Carry trade returns and volatility sorted on previous month's signals.



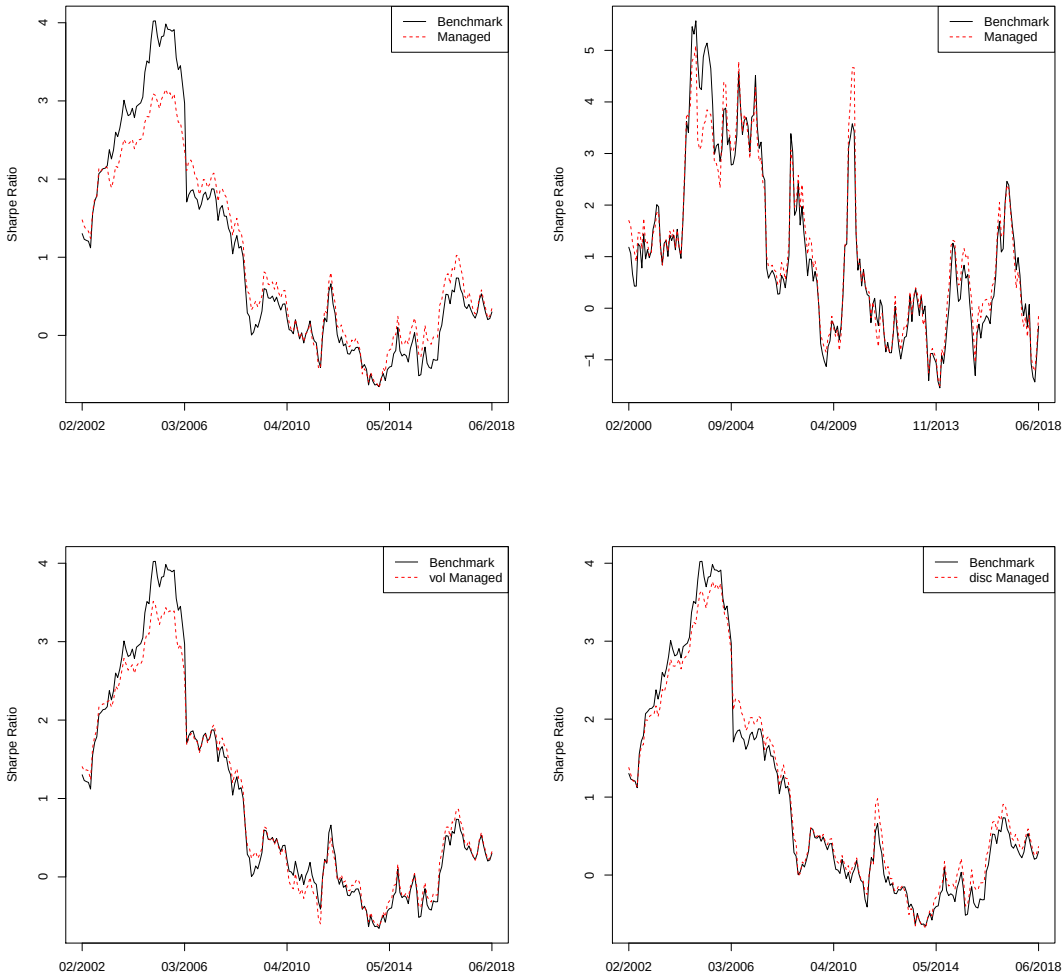
2.a Upper left graph: The annualised mean return of the benchmark carry trade sorted on previous month average forward discount. 2.b Upper right graph: The annualised mean return of the carry trade sorted on the previous month mean volatility of the global currency portfolio. 2.c Lower left graph: The volatility of the global currency portfolio sorted on the previous month's volatility of the global currency portfolio. 2.d Lower right graph: The annualised mean return of the carry trade sorted on the previous month's risk-adjusted return (RaR). For all graphs, the sample is from January 1999 to July 2018 for 34 currencies, as defined in Section 3. The carry trade portfolio is an equally weighted portfolio investing in the four highest-yielding currencies and selling the four lowest-yielding currencies. The currencies are ranked on their forward discount, and the portfolio is rebalanced every month. The global currency portfolio is an equally weighted portfolio including all currencies in the sample. We use the monthly time series of realized forward discount and volatility to sort the following month's returns and volatility into five buckets (quintiles). L represents the lowest bucket, while H is the highest bucket. The graphs show that the relationship between the forward discount and the future mean return is positive. It is also positive between volatility and future volatility. Altogether, the relationship is positive between the risk-adjusted expected return and future returns.

Figure 3: Time varying leverage and returns.



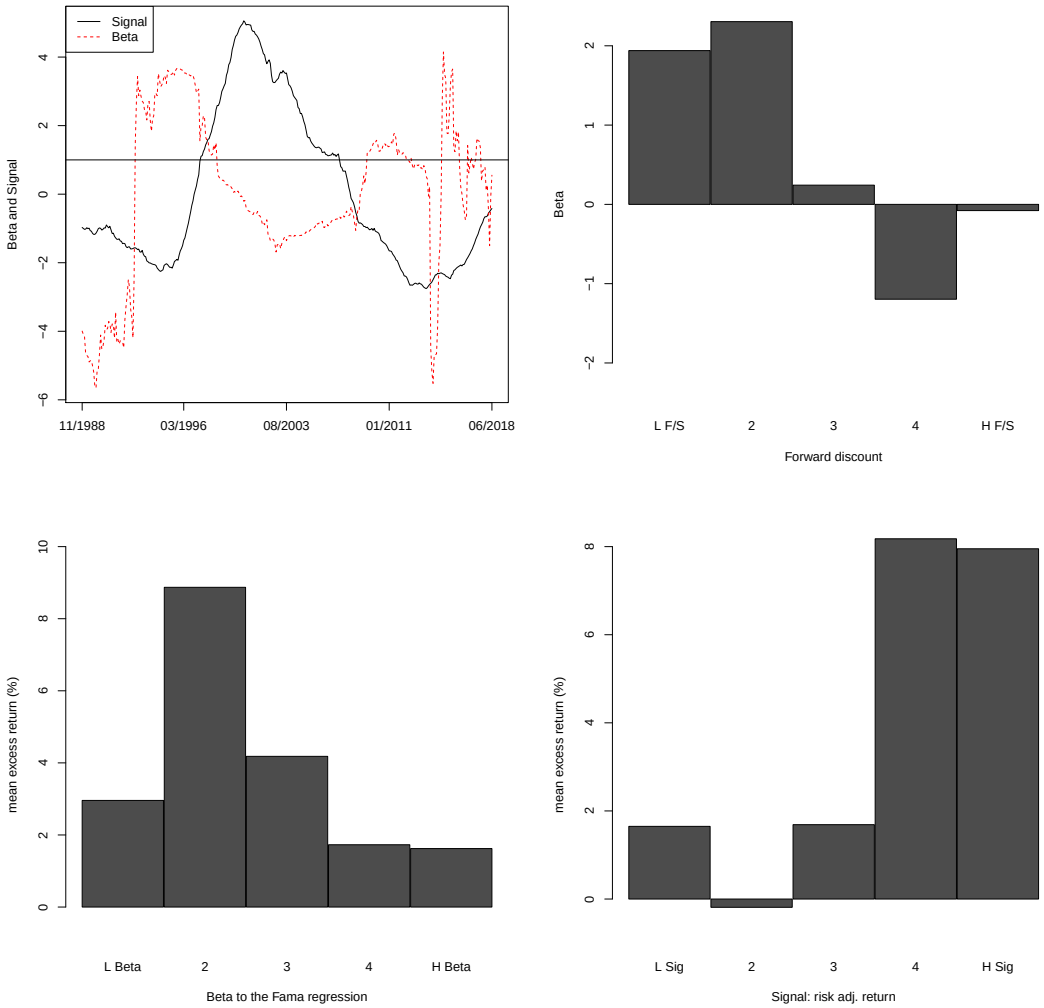
3.a Upper left graph: The ratio of the observed forward discount per currency to the observed volatility, as defined by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. The ratio defines the leverage of the risk-adjusted return-managed investment in the carry trade. 3.b Upper right graph: The volatility-adjusted cumulated returns of the managed strategy (solid line) and the benchmark strategy (dashed line). 3.c Lower left graph: The mean forward discount per currency of the highest-yielding currencies for which $F_b/S_a > 1$ minus the mean forward premium per currency of the four lowest-yielding currencies for which $F_a/S_b < 1$, as defined in equation (2). This series is also the numerator of Equation (4) $E(\pi_{t+1})$. Both time series have been multiplied by 1000 for ease of reading. 3.d Lower right graph: The observed volatility as defined in equation (3) of the equally weighted currency portfolio, including all currencies in the sample. This series is also the denominator of Equation (4) $\frac{1}{(MV_t)}$. For all graphs, the sample consists of the monthly observations from January 1999 to July 2018 for 34 currencies as defined in Section 3. The carry trade portfolio is an equally weighted portfolio investing in the four highest-yielding currencies and selling the four lowest-yielding currencies. The currencies are ranked on their forward discount, and the portfolio is rebalanced every month.

Figure 4: Rolling Sharpe ratio



Sample from January 1999 to July 2018 for 34 currencies as defined in Section 3. Monthly observations. 4.a Upper left graph: The annualized Sharpe Ratio calculated over a rolling window of 36 months. The benchmark strategy invests one dollar in a portfolio of long positions in the high-interest-rate currencies and one dollar in a portfolio of short positions in currencies with low interest rates. The strategy is rebalanced every month, and the proceeds are not reinvested. The portfolio is equally weighted and contains four currencies of each type. The managed strategy invests in the same portfolio but with time-varying leverage controlled by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. The graph shows that the managed strategy tends to overperform the benchmark strategy in both good and bad times. 4.b Upper right graph: The annualized Sharpe Ratio calculated over a rolling window of 12 months. 4.c Lower left graph: The annualized Sharpe Ratio calculated over a rolling window of 36 months. The benchmark strategy invests one dollar in a portfolio of long positions in high-interest-rate currencies and one dollar in a portfolio of short positions in currencies with low interest rates. The managed strategy invests in the same portfolio but with time-varying leverage controlled by $E(\pi_{t+1})$. 4.d Lower right graph: The annualized Sharpe Ratio calculated over a rolling window of 36 months. The benchmark strategy invests one dollar in a portfolio of long positions in high-interest-rate currencies and one dollar in a portfolio of short positions in currencies with low interest rates. The managed strategy invests in the same portfolio but with time-varying leverage controlled by $\frac{1}{(MV_t)}$.

Figure 5: Rolling Fama regression



Sample from November 1983 to July 2018 for 39 currencies as defined in Section 3. Monthly observations. 5.a Upper left graph: The five-year rolling beta of the Fama regression of the change in spot exchange rates on forward discounts. The regression is run on the portfolio containing the time varying set of four currencies with the largest forward discount to the dollar (dashed line). The graph also shows the five-year mean value of the ratio of the observed forward discount per currency to the observed volatility, as defined by Equation (4) $Signal_t = \frac{E(\pi_{t+1})}{(MV_t)}$. This ratio defines the leverage of the risk-adjusted-return-managed investment in the carry trade (solid line). 5.b Upper right graph: The five-year rolling beta of the Fama regression sorted on the average forward discount that is attractive to investors 5.c Lower left graph: The mean excess return of the augmented carry trade over the benchmark carry trade sorted on the beta of the five-year Fama regression. 5.d Lower right graph: The mean excess return of the augmented carry trade over the benchmark carry trade sorted on the value of the mean $Signal_t$.